

Chapter 6

IPv4 Addresses – Part 1

Rick Graziani
Cabrillo College



Number Systems

Digits (2): 0, 1

Number of:

	2^7				2^3	2^2	2^1	2^0
	<u>128's</u>	<u> </u>	<u> </u>	<u> </u>	<u>8's</u>	<u>4's</u>	<u>2's</u>	<u>1's</u>
Dec.								
2							1	0
10					1	0	1	0
17								
70								
130								
255								

Digits (2): 0, 1

Number of:

	2^7	2	2	2	2^3	2^2	2^1	2^0
	<u>128's</u>	<u>64's</u>	<u>32's</u>	<u>16's</u>	<u>8's</u>	<u>4's</u>	<u>2's</u>	<u>1's</u>
Dec.								
2							1	0
10					1	0	1	0
17				1	0	0	0	1
70		1	0	0	0	1	1	0
130	1	0	0	0	0	0	1	0
255	1	1	1	1	1	1	1	1

Digits (2): 0, 1

Number of:

	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
	<u>128's</u>	<u>64's</u>	<u>32's</u>	<u>16's</u>	<u>8's</u>	<u>4's</u>	<u>2's</u>	<u>1's</u>
Dec.		1	0	0	0	1	1	0
			1	0	1	0	0	0
	0	0	0	0	0	0	0	0
	1	0	0	0	0	0	0	0

172

192

Digits (2): 0, 1

Number of:

	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
	<u>128's</u>	<u>64's</u>	<u>32's</u>	<u>16's</u>	<u>8's</u>	<u>4's</u>	<u>2's</u>	<u>1's</u>
Dec.								
70		1	0	0	0	1	1	0
40			1	0	1	0	0	0
0	0	0	0	0	0	0	0	0
128	1	0	0	0	0	0	0	0
172	1	0	1	0	1	1	0	0
192	1	1	0	0	0	0	0	0

Binary to/from Decimal

Convert Decimal to Binary

172.16.4.20

Separate and convert each
decimal number separately

172

10101100

We start with the 172.

172 is greater than 128, place a 1 in the 128 position

- 128 and subtract 128

44 is less than 64, place a 0 in the 64 position

- 0

44 is greater than 32, place a 1 in the 32 position

- 32 and subtract 32

12 is less than 16, place a 0 in the 16 position

- 0

12 is greater than 8, place a 1 in the 8 position

- 8 and subtract 8

4 is equal to 4, place a 1 in the 4 position

- 4 and subtract 4

0 is less than 2, place a 0 in the 2 position

- 0

0 is less than 1, place a 0 in the 1 position

- 0

0 ALL DONE

Answer: 172 = 10101100

- Chapter 6 (Book and Curriculum) provides several methods and examples for doing the conversion between binary and decimal.

Rick Graziani
Cabrillo College

IPv4

Classful IP Addressing

Class A	Network	Host		
Octet	1	2	3	4

Class B	Network		Host	
Octet	1	2	3	4

Class C	Network			Host
Octet	1	2	3	4

Class D	Host			
Octet	1	2	3	4

Address Class	First Octet Range	Number of Possible Networks	Number of Hosts per Network
Class A	0 to 127	128 (2 are reserved)	16,777,214
Class B	128 to 191	16,348	65,534
Class C	192 to 223	2,097,152	254

- In the early days of the Internet, IP addresses were allocated to organizations based on request rather than actual need.
- When an organization received an IP network address, that address was associated with a “**Class**”, **A, B, or C**.
- This is known as **Classful IP Addressing**
- The **first octet** of the address determined what class the network belonged to and which bits were the network bits and which bits were the host bits.
- There were **no** subnet masks.
- It was not until 1992 when the IETF introduced CIDR (Classless Interdomain Routing), making the address class meaning less.
- This is known as **Classless IP Addressing**.

IPv4 Address Classes

Class A	Network	Host		
Octet	1	2	3	4

Class B	Network		Host	
Octet	1	2	3	4

Class C	Network			Host
Octet	1	2	3	4

Class D	Host			
Octet	1	2	3	4

Address Classes

	1st octet	2nd octet	3rd octet	4th octet
Class A	Network	Host	Host	Host
Class B	Network	Network	Host	Host
Class C	Network	Network	Network	Host

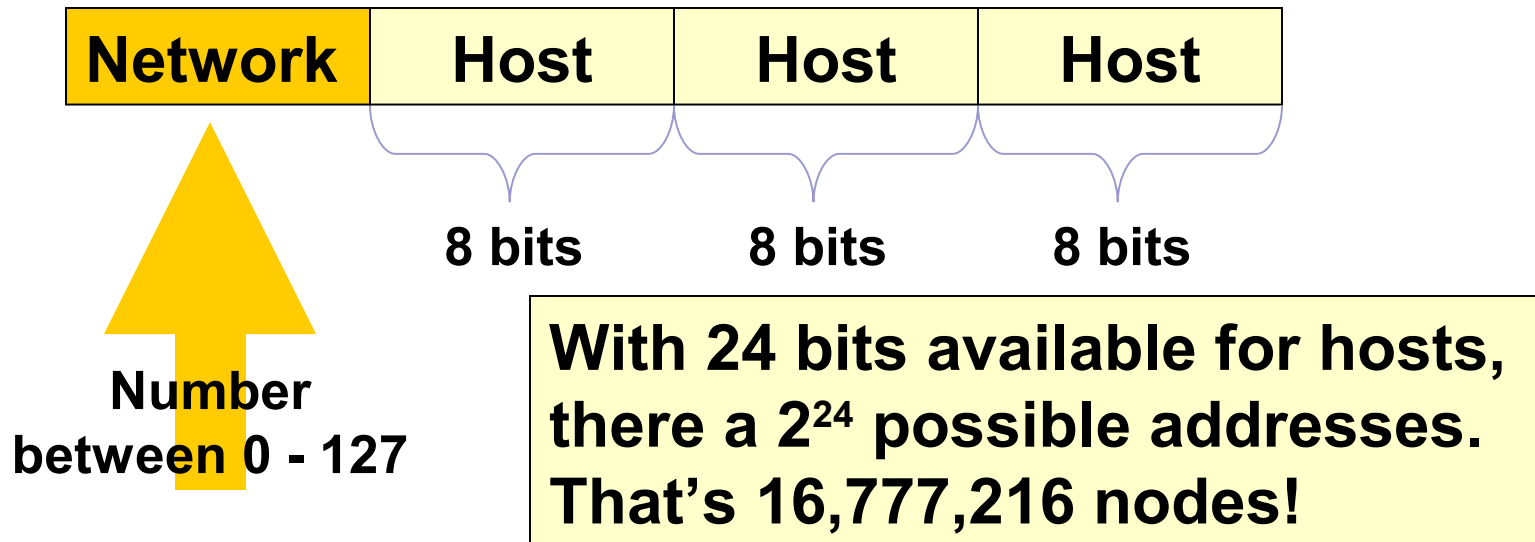
N = Network number

H = Host number

Class A addresses

Default Mask: 255.0.0.0 (/8)

First octet is between 0 – 127, begins with 0

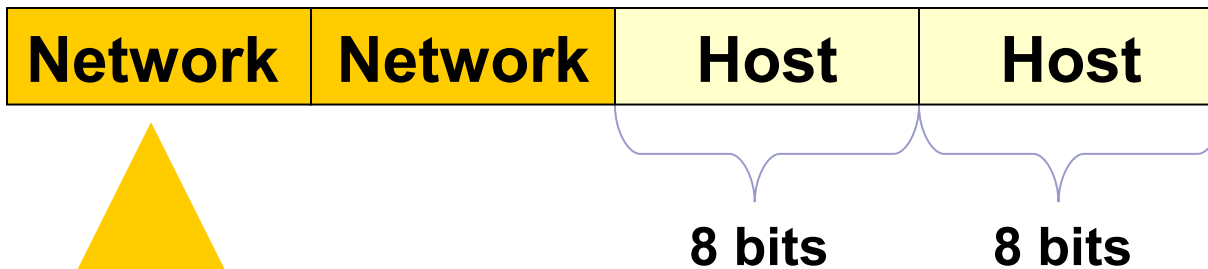


- There are 126 class A addresses.
 - 0 and 127 have special meaning and are not used.
- 16,777,214 host addresses, one for network address and one for broadcast address.
- Only large organizations such as the military, government agencies, universities, and large corporations have class A addresses.
- For example ISPs have 24.0.0.0 and 63.0.0.0
- Class A addresses account for 2,147,483,648 of the possible IPv4 addresses.

Default Mask: 255.255.0.0 (/16)

Class B addresses

First octet is between 128 – 191, begins with 10



**Number
between
128 - 191**

**With 16 bits available for hosts,
there a 2^{16} possible addresses.
That's 65,536 nodes!**

- There are 16,384 (2^{14}) class B networks.
- 65,534 host addresses, one for network address and one for broadcast address.
- Class B addresses are assigned to large organizations including corporations (such as Cisco, government agencies, and school districts).

Default Mask: 255.255.255.0 (/24)

Class C addresses

First octet is between 192 – 223, begins with 110

Network	Network	Network	Host
---------	---------	---------	------


**Number
between
192 - 223**

8 bits

**With 8 bits available for hosts,
there a 2^8 possible addresses.
That's 256 nodes!**

- There are 2,097,152 possible class C networks.
- 254 host addresses, one for network address and one for broadcast address.

IPv4 Address Classes

Class A	Network	Host		
Octet	1	2	3	4

Class B	Network		Host	
Octet	1	2	3	4

Class C	Network			Host
Octet	1	2	3	4

Class D	Host			
Octet	1	2	3	4

Address Class	First Octet Range	Number of Possible Networks	Number of Hosts per Network
Class A	0 to 127	128 (2 are reserved)	16,777,214
Class B	128 to 191	16,348	65,534
Class C	192 to 223	2,097,152	254

Network based on first octet

Address Class	First Octet Range	Number of Possible Networks	Number of Hosts per Network
Class A	0 to 127	128 (2 are reserved)	16,777,214
Class B	128 to 191	16,348	65,534
Class C	192 to 223	2,097,152	254

- The network portion of the IP address was dependent upon the first octet.
- There was no “Base Network Mask” provided by the ISP.
- The network mask was inherent in the address itself.

IPv4 Address Classes

Address Class	First Octet Range	Number of Possible Networks	Number of Hosts per Network
Class A	0 to 127	128 (2 are reserved)	16,777,214
Class B	128 to 191	16,348	65,534
Class C	192 to 223	2,097,152	254

Class D Addresses

- A Class D address begins with binary 1110 in the first octet.
- First octet range 224 to 239.
- Class D address can be used to represent a group of hosts called a host group, or multicast group.

Class E Addresses

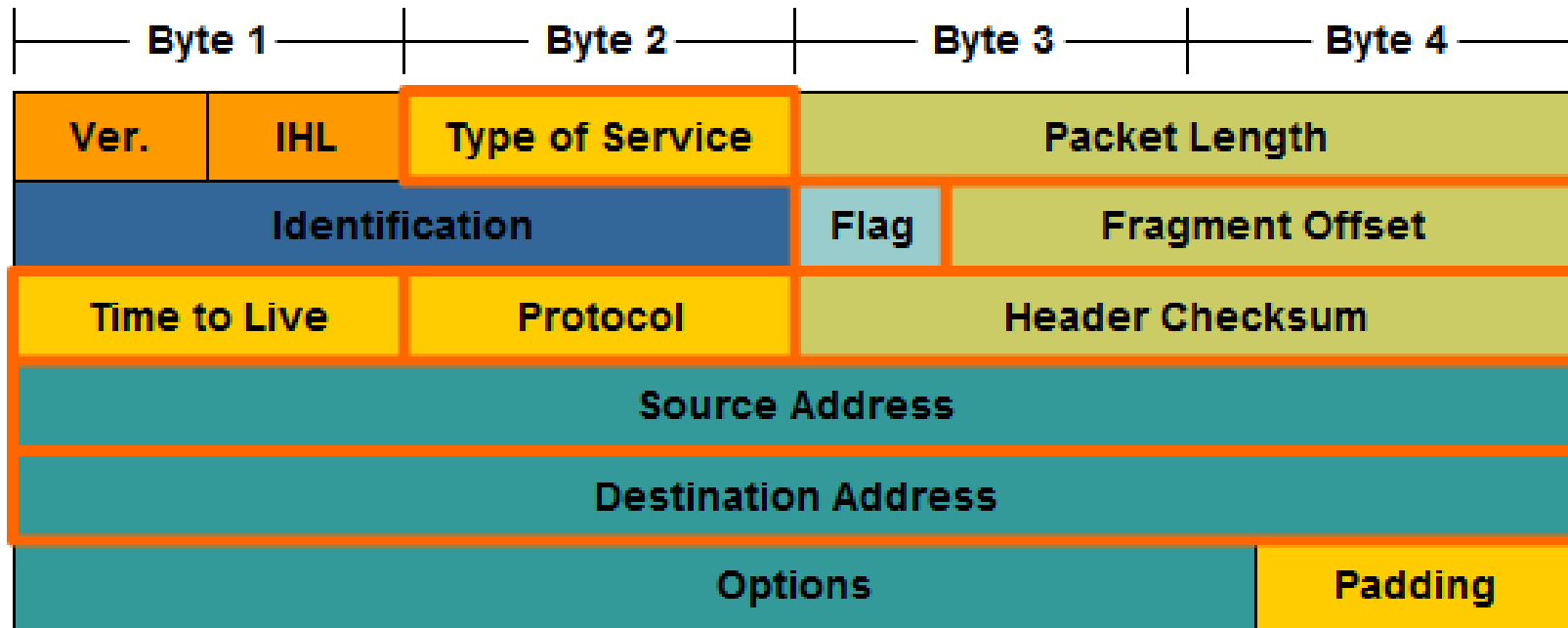
First octet of an IP address begins with 1111

- Class E addresses are reserved for experimental purposes and should not be used for addressing hosts or multicast groups.

Rick Graziani
Cabrillo College

IPv4 Addresses

IPv4 Addresses



- IPv4 addresses are 32 bit addresses

IPv4 Addresses

- IPv4 Addresses are 32 bit addresses:

1010100111000111010001011000100

10101001 11000111 01000101 10001001

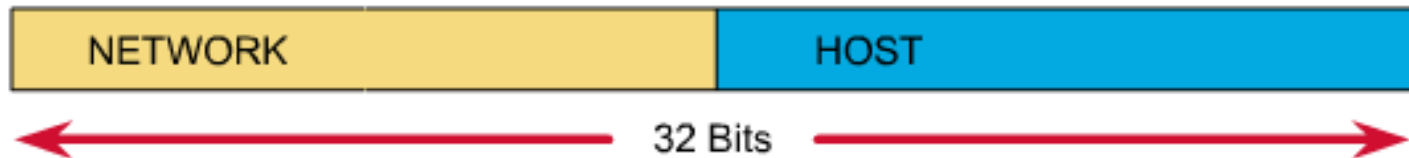
- We use **dotted notation** (or **dotted decimal notation**) to represent the value of each byte (octet) of the IP address in decimal.

10101001 11000111 01000101 10001001
169 . 199 . 69 . 137

IPv4 Addresses

An IP address has two parts:

- **network number**
- **host number**



Which bits refer to the network number?

Which bits refer to the host number?

IPv4 Addresses

Answer:

- Newer technology - **Classless IP Addressing**
 - The **subnet mask** determines the network portion and the host portion.
 - Value of first octet does NOT matter (older classful IP addressing)
 - Hosts and Classless Inter-Domain Routing (CIDR).
 - Classless IP Addressing is what is used within the Internet and in most internal networks.

- Older technology - **Classful IP Addressing (later)**
 - **Value of first octet** determines the network portion and the host portion.
 - Used with classful routing protocols like RIPv1.
 - The Cisco IP Routing Table is structured in a classful manner

Types of Addresses

Network Address

Broadcast Address

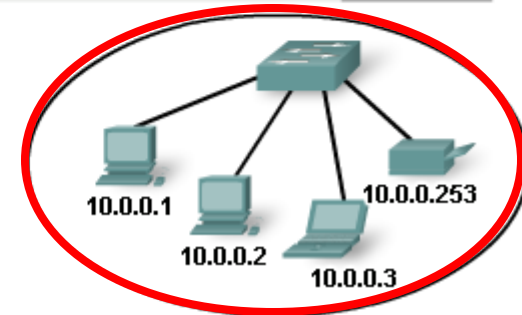
Host Address

Roll over to learn more.

Network Addresses have all 0's in the host portion.

Subnet Mask: 255.255.255.0

Network			Host
10	0	0	0
00001010	00000000	00000000	00000000
10	0	0	255
00001010	00000000	00000000	11111111
10	0	0	1
00001010	00000000	00000000	00000001



- **Network address** - The address by which we refer to the network
- **Broadcast address** - A special address used to send data to all hosts in the network
- **Host addresses** - The addresses assigned to the end devices in the network

Types of Addresses

Network Address

Broadcast Address

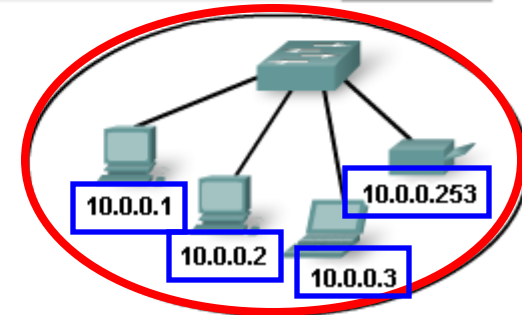
Host Address

Roll over to learn more.

Broadcast Addresses have all 1's in the host portion.

Subnet Mask: 255.255.255.0

Network			Host
10	0	0	0
00001010	00000000	00000000	00000000
10	0	0	255
00001010	00000000	00000000	11111111
10	0	0	1
00001010	00000000	00000000	00000001



- **Network address** - The address by which we refer to the network
- **Broadcast address** - A special address used to send data to all hosts in the network
- **Host addresses** - The addresses assigned to the end devices in the network

Types of Addresses

Network Address

Broadcast Address

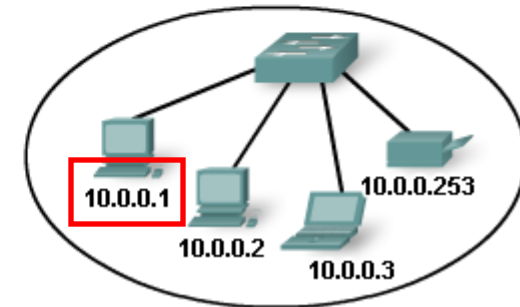
Host Address

Roll over to learn more.

Host Addresses can not have all 0's or all 1's in the host portion.

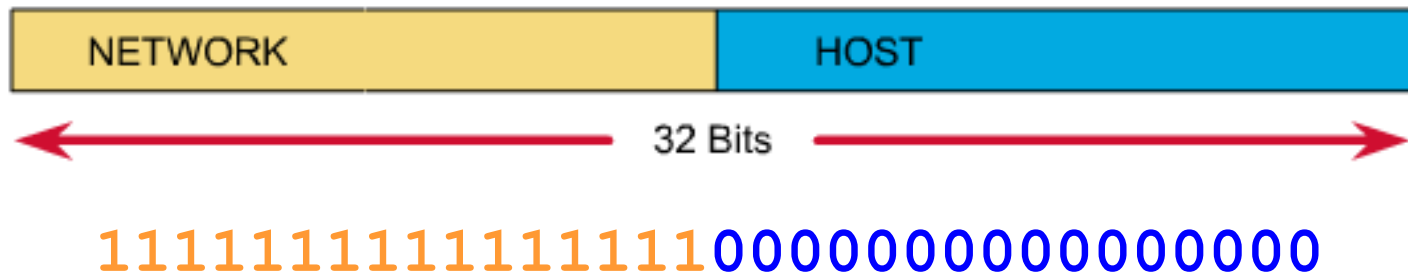
Subnet Mask: 255.255.255.0

Network			Host
10	0	0	0
00001010	00000000	00000000	00000000
10	0	0	255
00001010	00000000	00000000	11111111
10	0	0	1
00001010	00000000	00000000	00000001



- **Network address** - The address by which we refer to the network
- **Broadcast address** - A special address used to send data to all hosts in the network
- **Host addresses** - The addresses assigned to the end devices in the network

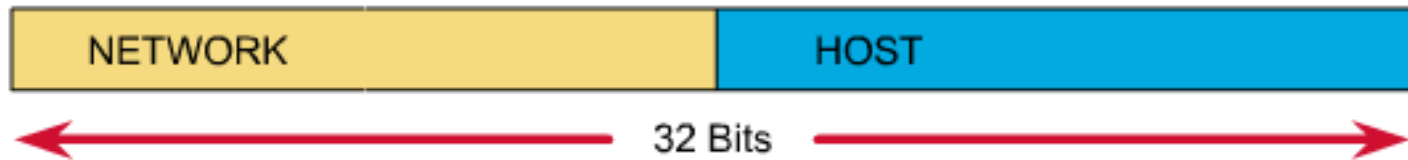
Dividing the Network and Host Portions



- **Subnet Mask**

- Used to define the:
 - Network portion
 - Host portion
- 32 bits
- Contiguous set of 1's followed by a contiguous set of 0's
 - 1's: Network portion
 - 0's: Host portion

Dividing the Network and Host Portions



11111111.11111111.00000000.00000000

Dotted decimal: 255 . 255 . 0 . 0

Slash notation: /16

- Expressed as:
 - Dotted decimal
 - Ex: 255.255.0.0
 - Slash notation or prefix length
 - /16 (the number of one bits)

Network Addresses

Network Address

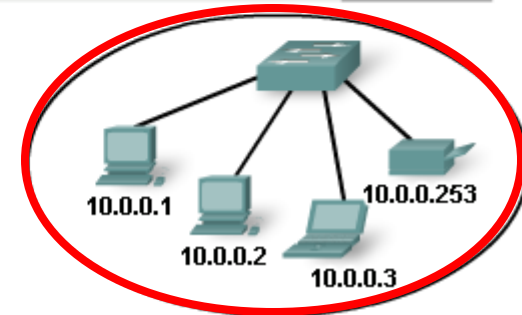
Broadcast Address

Host Address

Roll over to learn more.

Subnet Mask: 255.255.255.0

Network			Host
10	0	0	0
00001010	00000000	00000000	00000000
10	0	0	255
00001010	00000000	00000000	11111111
10	0	0	1
00001010	00000000	00000000	00000001



- **Network address** - The address by which we refer to the network
 - All binary 0's in the host portion of the address (more later)

Example 1

Network Address: 192.168.1.0

Subnet Mask: 255.255.255.0

192.168.1.0

Network Host

Network Address in binary:

11000000.10101000.00000001.00000000

Subnet Mask in binary:

11111111.11111111.11111111.00000000

Prefix Length: /24

Example 2

Network Address: 172.0.0.0

Subnet Mask: 255.0.0.0

172.0.0.0

Network Host

Network Address in binary:

10101100.00000000.00000000.00000000

Subnet Mask in binary:

11111111.00000000.00000000.00000000

Prefix Length : /8

Example 3

Network Address: 172.0.0.0

Subnet Mask: 255.255.0.0

172.0.0.0

Network Host

Network Address in binary:

10101100.00000000.00000000.00000000

Subnet Mask in binary:

11111111.11111111.00000000.00000000

Prefix Length: /16

Subnet Masks – Your Turn!

Underline the network portion of each address:

<u>Network Address</u>	<u>Subnet Mask</u>
172.0.0.0	255.0.0.0
172.16.0.0	255.255.0.0
192.168.1.0	255.255.255.0
192.168.0.0	255.255.0.0
192.168.0.0	255.255.255.0
10.1.1.0	/24
10.2.0.0	/16
10.0.0.0	/16

- What is the other portion of the address?

Subnet Masks – Your Turn!

Underline the network portion of each address:

<u>Network Address</u>	<u>Subnet Mask</u>
<u>172.0.0.0</u>	255.0.0.0
<u>172.16.0.0</u>	255.255.0.0
<u>192.168.1.0</u>	255.255.255.0
<u>192.168.0.0</u>	255.255.0.0
<u>192.168.0.0</u>	255.255.255.0
<u>10.1.1.0</u>	/24
<u>10.2.0.0</u>	/16
<u>10.0.0.0</u>	/16

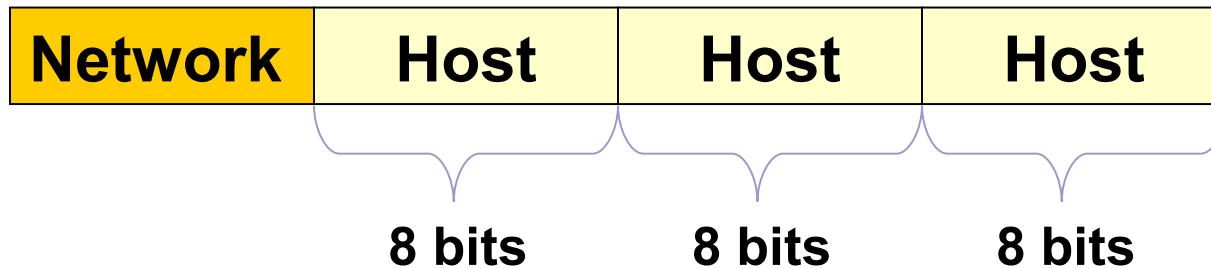
- What is the other portion of the address?
 - Host portion for host addresses

Why the mask matters: Number of hosts!

Subnet Mask:	1st octet	2nd octet	3rd octet	4th octet
255.0.0.0 or /8	Network	Host	Host	Host
255.255.0.0 or /16	Network	Network	Host	Host
255.255.255.0 or /24	Network	Network	Network	Host

- The more host bits in the subnet mask means the more hosts in the network.
- Subnet masks do not have to end on “natural octet boundaries”

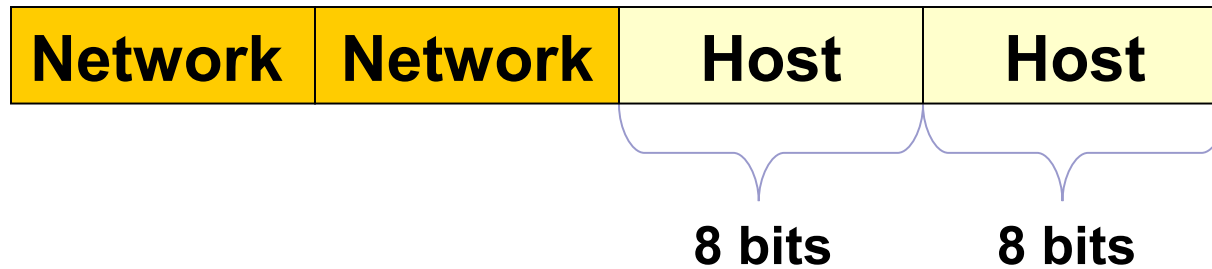
Subnet: 255.0.0.0 (/8)



**With 24 bits available for hosts,
there are 2^{24} possible addresses.
That's 16,777,216 nodes!**

- Only large organizations such as the military, government agencies, universities, and large corporations have networks with these many addresses.
- Example: A certain cable modem ISP has 24.0.0.0 and a DSL ISP has 63.0.0.0

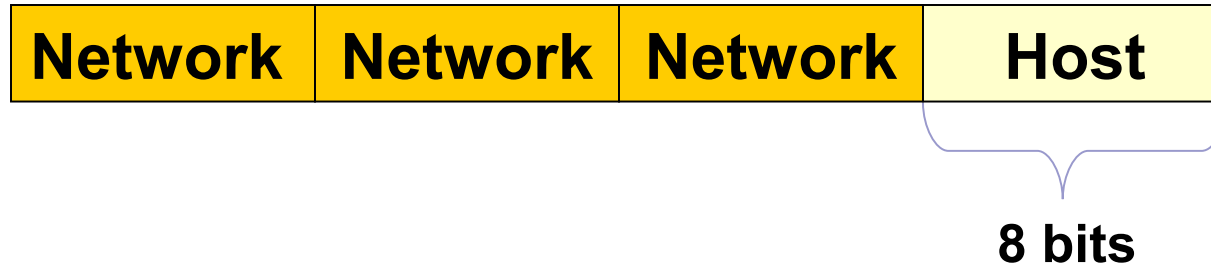
Subnet: 255.255.0.0 (/16)



**With 16 bits available for hosts,
there are 2^{16} possible addresses.
That's 65,536 nodes!**

- 65,534 host addresses, one for network address and one for broadcast address.

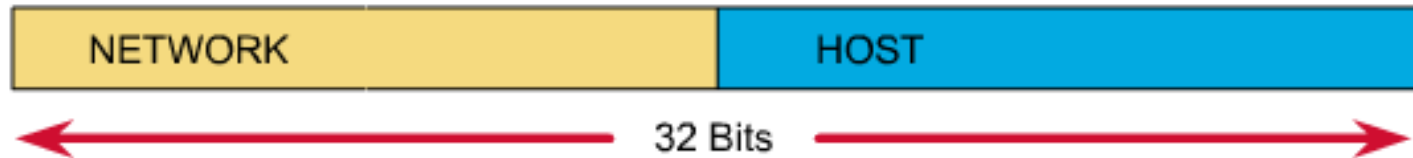
Subnet: 255.255.255.0 (/24)



**With 8 bits available for hosts,
there a 2^8 possible addresses.
That's 256 nodes!**

- 254 host addresses, one for network address and one for broadcast address.

IP Addresses



There is a tradeoff between:

- The number of network bits and the number of networks (subnets) you can have...

AND

- The number of HOST bits and the number of hosts for each network you can have.

Broadcast Addresses

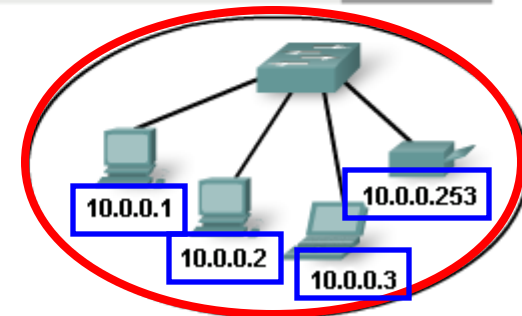
Network Address

Broadcast Address

Host Address

Roll over to learn more.

Network			Host
10	0	0	0
00001010	00000000	00000000	00000000
10	0	0	255
00001010	00000000	00000000	11111111
10	0	0	1
00001010	00000000	00000000	00000001



- **Broadcast address** - A special address used to send data to all hosts in the network
 - All binary 1's in the host portion of the address (more later)

Subnet Masks – Your Turn!

What is the broadcast address of each network:

<u>Network Address</u>	<u>Subnet Mask</u>	<u>Broadcast Address</u>
172.0.0.0	255.0.0.0	
172.16.0.0	255.255.0.0	
192.168.1.0	255.255.255.0	
192.168.0.0	255.255.0.0	
192.168.0.0	255.255.255.0	
10.1.1.0	/24	
10.2.0.0	/16	
10.0.0.0	/16	

Subnet Masks – Your Turn!

What is the broadcast address of each network:

<u>Network Address</u>	<u>Subnet Mask</u>	<u>Broadcast Address</u>
172.0.0.0	255.0.0.0	172.255.255.255
172.16.0.0	255.255.0.0	172.16.255.255
192.168.1.0	255.255.255.0	192.168.1.255
192.168.0.0	255.255.0.0	192.168.255.255
192.168.0.0	255.255.255.0	192.168.0.255
10.1.1.0	/24	10.1.1.255
10.2.0.0	/16	10.2.255.255
10.0.0.0	/16	10.0.255.255



Bringing it all together

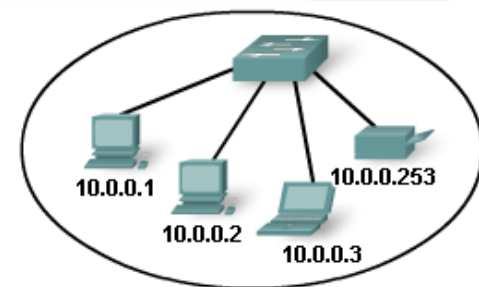
Network Address

Broadcast Address

Host Address

Roll over to learn more.

Network			Host
10	0	0	0
00001010	00000000	00000000	00000000
10	0	0	255
00001010	00000000	00000000	11111111
10	0	0	1
00001010	00000000	00000000	00000001



- Subnet Mask divides Network portion and Host portion:
 - 1's: Network portion
 - 0's: Host portion
- Network address:
 - All 0's in the host portion of the address
- Broadcast address:
 - All 1's in the host portion of the address

Bringing it all together

- Convert these addresses and masks to Binary (to be used later)

Network: 172.0.0.0	_____	.	_____	.	_____	.	_____
Mask: 255.0.0.0	_____	.	_____	.	_____	.	_____
172.255.255.255	_____	.	_____	.	_____	.	_____
Broadcast Address							

Network: 172.16.0.0	_____	.	_____	.	_____	.	_____
Mask: 255.255.0.0	_____	.	_____	.	_____	.	_____
172.16.255.255	_____	.	_____	.	_____	.	_____
Broadcast Address							

Bringing it all together

- Convert these addresses and masks to Binary (to be used later)

Network: 172.0.0.0 10101100.00000000.00000000.00000000
Mask: 255.0.0.0 11111111.00000000.00000000.00000000
 172.255.255.255 10101100.11111111.11111111.11111111
Broadcast Address

Network: 172.16.0.0 10101100.00010000.00000000.00000000
Mask: 255.255.0.0 11111111.11111111.00000000.00000000
172.16.255.255 10101100.00010000.11111111.11111111
Broadcast Address

Bringing it all together

- Convert these addresses and masks to Binary (to be used later)

Network: 192.168.1.0	_____	.	_____	.	_____	.	_____
Mask: 255.255.255.0	_____	.	_____	.	_____	.	_____
Bcst: 192.168.1.255	_____	.	_____	.	_____	.	_____

Network: 192.168.0.0	_____	.	_____	.	_____	.	_____
Mask: 255.255.0.0	_____	.	_____	.	_____	.	_____
Bcst: 192.168.255.255	_____	.	_____	.	_____	.	_____

Network: 192.168.0.0	_____	.	_____	.	_____	.	_____
Mask: 255.255.255.0	_____	.	_____	.	_____	.	_____
Bcst: 192.168.0.255	_____	.	_____	.	_____	.	_____

Bringing it all together

- Convert these addresses and masks to Binary (to be used later)

Network:	192.168.1.0	11000000.10101000.00000001.00000000
Mask:	255.255.255.0	11111111.11111111.11111111.00000000
Bcst:	192.168.1.255	11000000.10101000.00000001.11111111

Network:	192.168.0.0	11000000.10101000.00000000.00000000
Mask:	255.255.0.0	11111111.11111111.00000000.00000000
Bcst:	192.168.255.255	11000000.10101000.11111111.11111111

Network:	192.168.0.0	11000000.10101000.00000000.00000000
Mask:	255.255.255.0	11111111.11111111.11111111.00000000
Bcst:	192.168.0.255	11000000.10101000.00000000.11111111

Bringing it all together

- Convert these addresses and masks to Binary (to be used later)

Network: 10.1.1.0 _____ . _____ . _____ . _____

Mask: /24 _____ . _____ . _____ . _____

Bcast: 10.1.1.255 _____ . _____ . _____ . _____

Network: 10.2.0.0 _____ . _____ . _____ . _____

Mask: /16 _____ . _____ . _____ . _____

Bst:10.2.255.255 _____ . _____ . _____ . _____

Network 10.0.0.0 _____ . _____ . _____ . _____

Mask: /16 _____ . _____ . _____ . _____

Bcast10.0.255.255 _____ . _____ . _____ . _____

Bringing it all together

- Convert these addresses and masks to Binary (to be used later)

Network: 10.1.1.0	00001010.00000001.00000001.00000000
Mask: /24	11111111.11111111.11111111.00000000
Bcast: 10.1.1.255	00001010.00000001.00000001.11111111

Network: 10.2.0.0	00001010.00000010.00000000.00000000
Mask: /16	11111111.11111111.00000000.00000000
Bst:10.2.255.255	00001010.00000010.11111111.11111111

Network 10.0.0.0	00001010.00000000.00000000.00000000
Mask: /16	11111111.11111111.00000000.00000000
Bcast10.0.255.255	00001010.00000000.11111111.11111111

Host IP Addresses



192.168.10.100/24

- Host IP Addresses contain:
 - Network portion of the address
 - Unique combination of 0's and 1's in the host portion of the address
 - Cannot be all 0's (network address)
 - Cannot be all 1's (broadcast address)
- Hosts have subnet masks to determine network portion (later)

Range of hosts – Your Turn!

- Host addresses are all addresses between the network address and the broadcast address.
- What is the range of host addresses for each network?

<u>Network Address</u>	<u>Subnet Mask</u>	<u>Broadcast Address</u>
172.0.0.0	255.0.0.0	172.255.255.255
172.16.0.0	255.255.0.0	172.16.255.255
192.168.1.0	255.255.255.0	192.168.1.255
192.168.0.0	255.255.0.0	192.168.255.255
192.168.0.0	255.255.255.0	192.168.0.255
10.1.1.0	/24	10.1.1.255
10.2.0.0	/16	10.2.255.255
10.0.0.0	/16	10.0.255.255

Range of hosts – Your Turn!

<u>Network Address</u>	<u>Subnet Mask</u>	<u>Broadcast Address</u>
172.0.0.0	255.0.0.0	172.255.255.255
172.0.0.1 through 172.255.255.254		
172.16.0.0	255.255.0.0	172.16.255.255
172.16.0.1 through 172.16.255.254		
192.168.1.0	255.255.255.0	192.168.1.255
192.168.1.1 through 192.168.1.254		
192.168.0.0	255.255.0.0	192.168.255.255
192.168.0.1 through 192.168.255.254		
192.168.0.0	255.255.255.0	192.168.0.255
192.168.0.1 through 192.168.0.254		

Range of hosts – Your Turn!

<u>Network Address</u>	<u>Subnet Mask</u>	<u>Broadcast Address</u>
10.1.1.0	/24	10.1.1.255
10.1.1.1 through 10.1.1.254		
10.2.0.0	/16	10.2.255.255
10.2.0.1 through 10.2.255.254		
10.0.0.0	/16	10.0.255.255
10.0.0.1 through 10.0.255.254		

Range of hosts – Your Turn!

- Host Addresses in binary

172.0.0.0 (net)	10101100.00000000.00000000.00000000
255.0.0.0 (SM)	11111111.00000000.00000000.00000000
172.0.0.1	_____.
172.255.255.254	_____.
172.255.255.255 (broadcast)	10101100.11111111.11111111.11111111

172.16.0.0 (net)	10101100.00010000.00000000.00000000
255.255.0.0 (SM)	11111111.11111111.00000000.00000000
172.16.0.1	_____.
172.16.255.254	_____.
172.16.255.255 (broadcast)	10101100.00010000.11111111.11111111

Range of hosts – Your Turn!

- Host Addresses in binary

172.0.0.0 (net)	10101100.00000000.00000000.00000000
255.0.0.0 (SM)	11111111.00000000.00000000.00000000
172.0.0.1	10101100.00000000.00000000.00000001
172.255.255.254	10101100.11111111.11111111.11111110
172.255.255.255 (broadcast)	10101100.11111111.11111111.11111111

172.16.0.0 (net)	10101100.00010000.00000000.00000000
255.255.0.0 (SM)	11111111.11111111.00000000.00000000
172.16.0.1	10101100.00010000.00000000.00000001
172.16.255.254	10101100.00010000.11111111.11111110
172.16.255.255 (broadcast)	10101100.00010000.11111111.11111111

Range of hosts – Your Turn!

- Host Addresses in binary

192.168.1.0 (net)	11000000.10101000.00000001.00000000
255.255.255.0 (SM)	11111111.11111111.11111111.00000000
192.168.1.1	_____.
192.168.1.254	_____.
192.168.1.255	11000000.10101000.00000001.11111111
(broadcast)	

192.168.0.0 (net)	11000000.10101000.00000000.00000000
255.255.0.0 (SM)	11111111.11111111.00000000.00000000
192.168.0.1	_____.
192.168.255.254	_____.
192.168.255.255	11000000.10101000.11111111.11111111
(broadcast)	

Range of hosts – Your Turn!

- Host Addresses in binary

192.168.1.0 (net)	11000000.10101000.00000001.00000000
255.255.255.0 (SM)	11111111.11111111.11111111.00000000
192.168.1.1	11000000.10101000.00000001.00000001
192.168.1.254	11000000.10101000.00000001.11111110
192.168.1.255 (broadcast)	11000000.10101000.00000001.11111111

192.168.0.0 (net)	11000000.10101000.00000000.00000000
255.255.0.0 (SM)	11111111.11111111.00000000.00000000
192.168.0.1	11000000.10101000.00000000.00000001
192.168.255.254	11000000.10101000.11111111.11111110
192.168.255.255 (broadcast)	11000000.10101000.11111111.11111111

Range of hosts – Your Turn!

- Host Addresses in binary

192.168.0.0 (net)	11000000.10101000.00000000.00000000
255.255.255.0 (SM)	11111111.11111111.11111111.00000000
192.168.0.1	_____.
192.168.0.254	_____.
192.168.0.255	11000000.10101000.00000000.11111111
(broadcast)	

Range of hosts – Your Turn!

- Host Addresses in binary

192.168.0.0 (net)	11000000.10101000.00000000.00000000
255.255.255.0 (SM)	11111111.11111111.11111111.00000000
192.168.0.1	11000000.10101000.00000000.00000001
192.168.0.254	11000000.10101000.00000000.11111110
192.168.0.255 (broadcast)	11000000.10101000.00000000.11111111

Range of hosts – The rest...

Host Addresses in binary

10.1.1.0 (net)	00001010.00000001.00000001.00000000
/24 (SM)	11111111.11111111.11111111.00000000
10.1.1.1	00001010.00000001.00000001.00000001
10.1.1.254	00001010.00000001.00000001.11111110
10.1.1.255 (broadcast)	00001010.00000001.00000001.11111111
10.2.0.0 (net)	00001010.00000010.00000000.00000000
/16 (SM)	11111111.11111111.00000000.00000000
10.2.0.1	00001010.00000010.00000000.00000001
10.2.255.254	00001010.00000010.11111111.11111110
10.2.255.255 (broadcast)	00001010.00000010.11111111.11111111

Range of hosts – The rest...

- Host Addresses in binary

10.0.0.0 (net)	00001010.00000000.00000000.00000000
/16 (SM)	11111111.11111111.00000000.00000000
10.0.0.1	00001010.00000000.00000000.00000001
10.0.255.254	00001010.00000000.11111111.11111110
10.0.255.255 (broadcast)	00001010.00000000.11111111.11111111

Subnet Masks: Non-Natural Boundaries

- Subnet masks do not have to end on natural octet boundaries
- Convert these to binary:

<u>Network Address</u>	<u>Subnet Mask</u>
172.1.16.0	255.255.240.0
192.168.1.0	255.255.255.224

Subnet Masks: Non-Natural Boundaries

- Subnet masks do not have to end on natural octet boundaries

172.1.16.0	10101100.00000001.00010000.00000000
255.255.240.0	11111111.11111111.11110000.00000000

- What is the range of host addresses in dotted-decimal and binary?
- What is the broadcast address?
- How many host addresses?

Subnet Masks: Non-Natural Boundaries

- Subnet masks do not have to end on natural octet boundaries

172.1.16.0 10101100.00000001.00010000.00000000

255.255.240.0 11111111.11111111.11110000.00000000

172.1.16.1 10101100.00000001.00010000.00000001

172.1.16.2 10101100.00000001.00010000.00000010

172.1.16.3 10101100.00000001.00010000.00000011

...

172.1.16.255 10101100.00000001.00010000.11111111

172.1.17.0 10101100.00000001.00010001.00000000

172.1.17.1 10101100.00000001.00010001.00000001

...

172.1.31.254 10101100.00000001.00011111.11111110

Subnet Masks: Non-Natural Boundaries

- Subnet masks do not have to end on natural octet boundaries

172.1.16.0 10101100.00000001.00010000.00000000

255.255.240.0 11111111.11111111.11110000.00000000

172.1.16.1 10101100.00000001.00010000.00000001

...

172.1.31.254 10101100.00000001.00011111.11111110

172.1.31.255 10101100.00000001.00011111.11111111
(broadcast)

Number of hosts: $2^{12} - 2 = 4,096 - 2 = 4,094$ hosts

Subnet Masks: Non-Natural Boundaries

- Subnet masks do not have to end on natural octet boundaries

192.168.1.0	11000000.10101000.00000001.00000000
255.255.255.224	11111111.11111111.11111111.11100000

192.168.1.1	11000000.10101000.00000001.00000001
192.168.1.2	11000000.10101000.00000001.00000010
192.168.1.3	11000000.10101000.00000001.00000011

...

192.168.1.29	11000000.10101000.00000001.00011101
192.168.1.30	11000000.10101000.00000001.00011110

192.168.1.31 (broadcast)	11000000.10101000.00000001.00011111
-----------------------------	-------------------------------------

Subnet Masks: Non-Natural Boundaries

- Subnet masks do not have to end on natural octet boundaries

192.168.1.0	11000000.10101000.00000001.00000000
255.255.255.224	11111111.11111111.11111111.11100000

192.168.1.1	11000000.10101000.00000001.00000001
-------------	-------------------------------------

...

192.168.1.30	11000000.10101000.00000001.00011110
--------------	-------------------------------------

192.168.1.31 (broadcast)	11000000.10101000.00000001.00011111
-----------------------------	-------------------------------------

Number of hosts: $2^5 - 2 = 32 - 2 = 30$ hosts

Chapter 6

IPv4 Addresses – Part 2

Rick Graziani
Cabrillo College

Host IP Addresses

Host IP Addresses

172.0.0.0 (net) 10101100.00000000.00000000.00000000

255.0.0.0 (SM) 11111111.00000000.00000000.00000000

172.0.0.1 10101100.00000000.00000000.00000001

172.255.255.254 10101100.11111111.11111111.11111110

172.255.255.255 10101100.11111111.11111111.11111111

(broadcast)

172.16.0.0 (net) 10101100.00010000.00000000.00000000

255.255.0.0 (SM) 11111111.11111111.00000000.00000000

172.16.0.1 10101100.00010000.00000000.00000001

172.16.255.254 10101100.00010000.11111111.11111110

172.16.255.255 10101100.00010000.11111111.11111111

(broadcast)

Host IP Addresses

172.1.16.0 10101100.00000001.00010000.00000000

255.255.240.0 11111111.11111111.11110000.00000000

172.1.16.1 10101100.00000001.00010000.00000001

...

172.1.31.254 10101100.00000001.00011111.11111110

172.1.31.255 10101100.00000001.00011111.11111111

(broadcast)

Number of hosts: $2^{12} - 2 = 4,096 - 2 = 4,094$ hosts

Who assigns IP Network Addresses?



- **Internet Assigned Numbers Authority (IANA)**
(<http://www.iana.net>) is the master holder of the IP addresses.
- Today, the remaining IPv4 address space has been allocated to various other registries to manage for particular purposes or for regional areas.
 - **Regional Internet Registries (RIRs)**

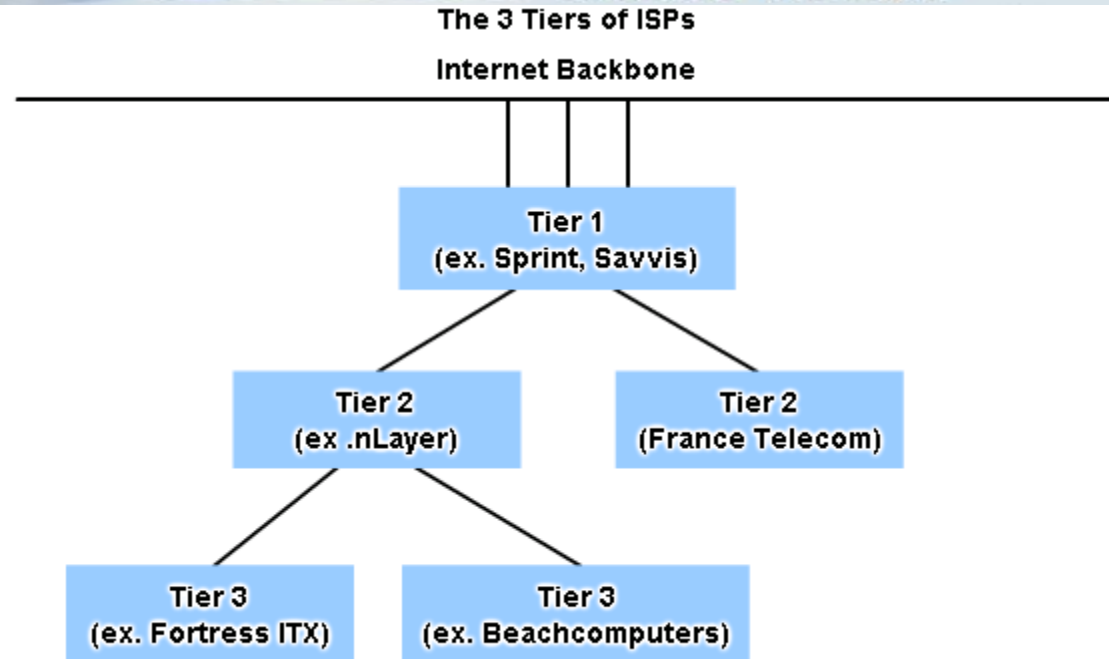
Regional Internet Registries (RIR)



- The 5 RIR's are:
 - **AfrINIC** (African Network Information Centre) - Africa Region
<http://www.afrinic.net>
 - **APNIC** (Asia Pacific Network Information Centre) - Asia/Pacific Region <http://www.apnic.net>
 - **ARIN** (American Registry for Internet Numbers) - North America Region
<http://www.arin.net>
 - **LACNIC** (Regional Latin-American and Caribbean IP Address Registry) - Latin America and some Caribbean Islands <http://www.lacnic.net>
 - **RIPE NCC** (Reseaux IP Europeans) - Europe, the Middle East, and Central Asia <http://www.ripe.net>

ISP (Internet Service Providers)

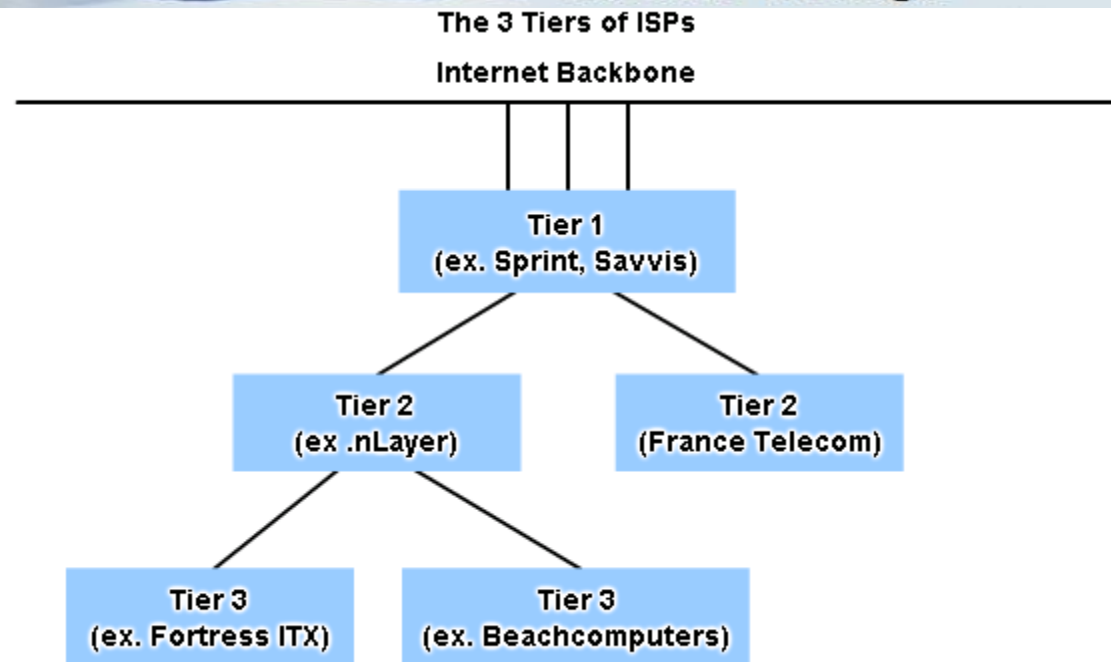
Most companies or organizations obtain their IPv4 address blocks from an ISP.



- Tier 1 ISP:
 - Large national or international ISPs that are directly connected to the Internet backbone.
 - Customers of Tier 1 ISPs:
 - lower-tiered ISPs
 - large companies and organizations.
 - Offer reliability and speed
 - AOL, SPRINT, Global Crossing, AT&T, Level 3, Verizon, NTT, Quest, SAVVIS

ISP (Internet Service Providers)

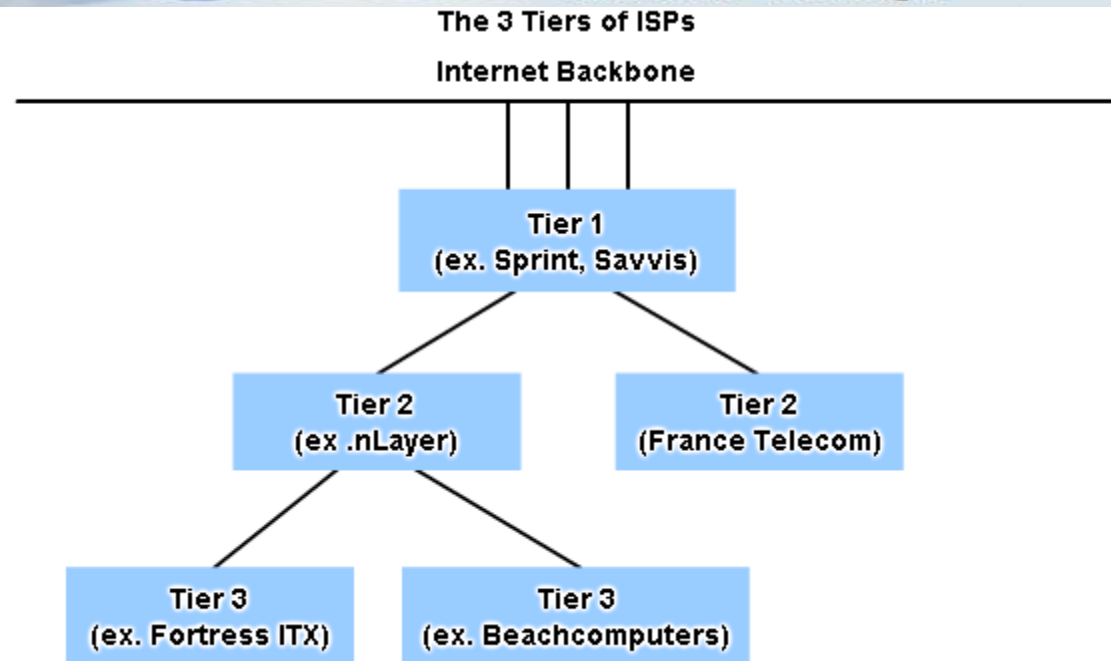
Most companies or organizations obtain their IPv4 address blocks from an ISP.



- Tier 2 ISP:
 - Acquire their Internet service from Tier 1 ISPs. Tier 2 ISPs generally focus on business customers.
 - Examples: Allstream, AboveNet, British Telecom, Cogent Communications, France Telecom, Teleglobe TeliaSonera International Carrier Time Warner Telecom, Tiscali International Network, XO Communications

ISP (Internet Service Providers)

Most companies or organizations obtain their IPv4 address blocks from an ISP.



- Tier 3 ISP:
 - Purchase their Internet service from Tier 2 ISPs. The focus of these ISPs is the retail and home markets in a specific locale. Examples:
 - Local ISPs

Special Unicast IPv4 Addresses

☒ Use the following IP address:

IP address: 192 . 168 . 1 . 100

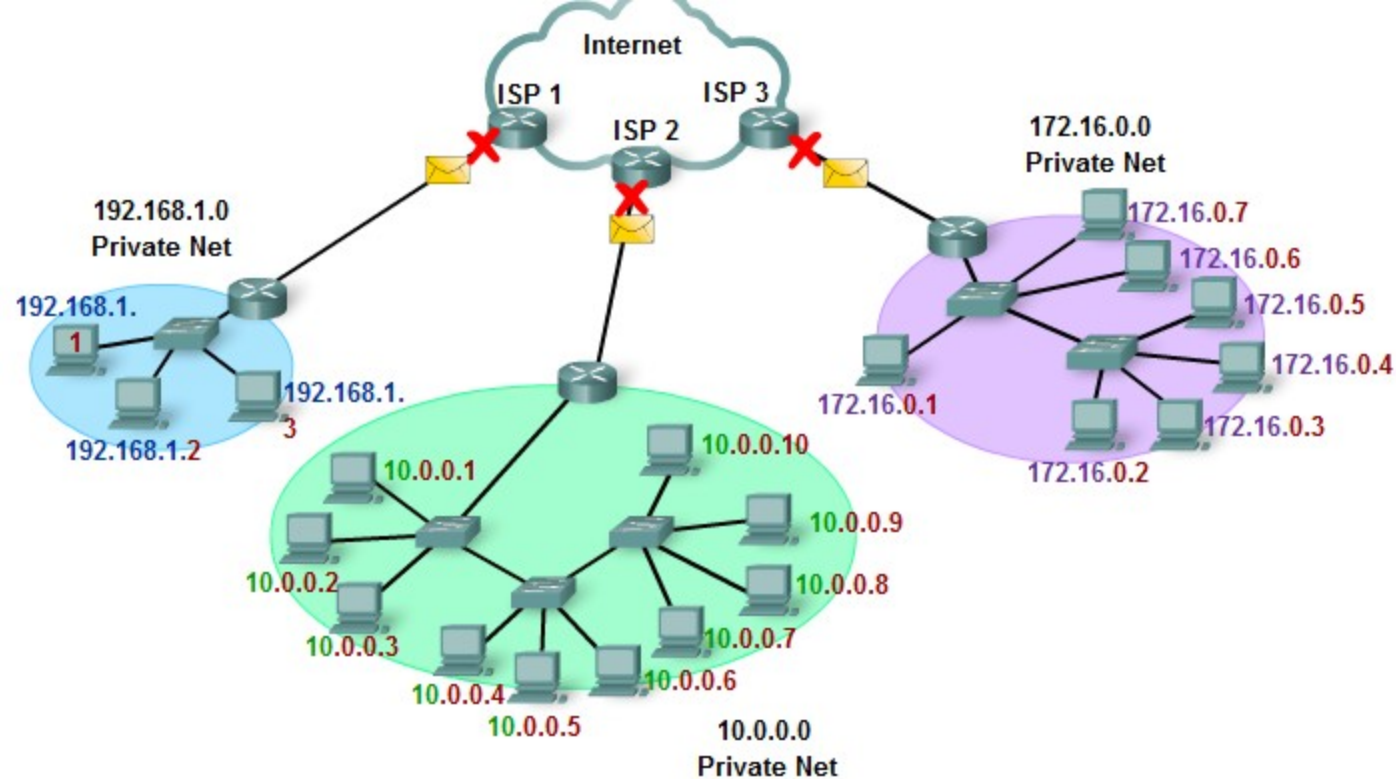
Subnet mask: 255 . 255 . 255 . 0

Default gateway: 192 . 168 . 1 . 1

- Default Route
- Loopback Address
 - Special address that hosts use to direct traffic to themselves.
 - 127.0.0.0 to 127.255.255.255
- Link-Local Addresses
 - 169.254.0.0 to 169.254.255.255 (169.254.0.0 /16)
 - Can be automatically assigned to the local host by the operating system in environments where no IP configuration is available.
- TEST-NET Addresses
 - 192.0.2.0 to 192.0.2.255 (192.0.2.0 /24)
 - Set aside for teaching and learning purposes.
 - These addresses can be used in documentation and network examples.



Private IP Addresses



- RFC 1918
 - 10.0.0.0 to 10.255.255.255 (10.0.0.0 /8)
 - 172.16.0.0 to 172.31.255.255 (172.16.0.0 /12)
 - 192.168.0.0 to 192.168.255.255 (192.168.0.0 /16)
- The addresses will not be routed in the Internet
 - Need NAT/PAT
- Should be blocked by your ISP
- Allows for any network to have up to 16,777,216 hosts (/8)

Rick Graziani
Cabrillo College

The Subnet Mask and the AND Operation

Subnet Mask

☒ Use the following IP address:

IP address:	192 . 168 . 1 . 100
Subnet mask:	255 . 255 . 255 . 0
Default gateway:	192 . 168 . 1 . 1

Host: "I'm a host on the 192.168.1.0/24 network."

- The subnet mask is used to separate the network portion from the host portion of the address.
- On a host, the subnet mask tells the host what network it belongs to.
- Why does a host need to know what network it belongs to?
- So, it knows whether to encapsulate the IP packet into an Ethernet frame with:
 - The Destination MAC Address of the default gateway
 - Must know the default gateway's IP address
 - The Destination MAC Address of the host with the Destination IP address of the packet

Subnet Mask

	Network	Host
Host IP: 172.16.33.10	10101100.00010000	.00100001.00001010
Mask: 255.255.0.0	11111111.11111111	.00000000.00000000

Net Add: 172.16.0.0	10101100.00010000	.00000000.00000000

- Devices such as hosts use the bit-wise AND operation on the:
 - Host IP address
 - Subnet mask
- AND operation:
 - 1 AND 1 = 1
 - 0 AND anything = 0

Subnet Mask

	Network	Host
Host IP: 172.16.33.10	10101100.00010000.00100001	00001010
Mask: 255.255.255.0	11111111.11111111.11111111	00000000

Net Add: 172.16.33.0	10101100.00010000.00100001	00000000

- AND operation:
 - 1 AND 1 = 1
 - 0 AND anything = 0

Subnet Mask

	Network	Host
Host IP: 172.1.17.9	10101100.00000001.00010001.00001001	
Mask: 255.255.240.0	11111111.11111111.11110000.00000000	
	-----	-----
Net Add: 172.1.16.0	10101100.00000001.00010000.00000000	

- AND operation:
 - 1 AND 1 = 1
 - 0 AND anything = 0

Subnet Masks: Non-Natural Boundaries

- Subnet masks do not have to end on natural octet boundaries

172.1.16.0 10101100.00000001.00010000.00000000

255.255.240.0 11111111.11111111.11110000.00000000

172.1.16.1 10101100.00000001.00010000.00000001

...

172.1.31.254 10101100.00000001.00011111.11111110

172.1.31.255 10101100.00000001.00011111.11111111
(broadcast)

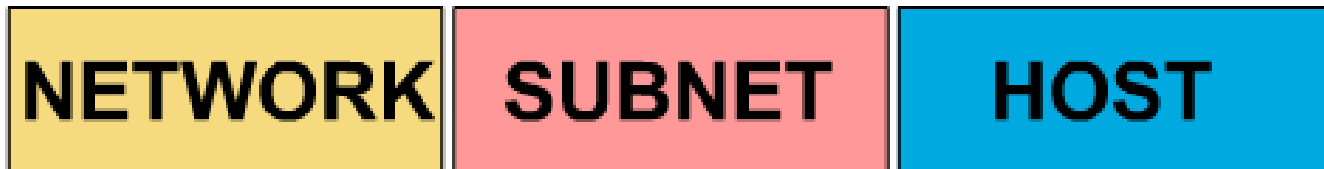
Number of hosts: $2^{12} - 2 = 4,096 - 2 = 4,094$ hosts

Rick Graziani
Cabrillo College



Subnetting: First Look

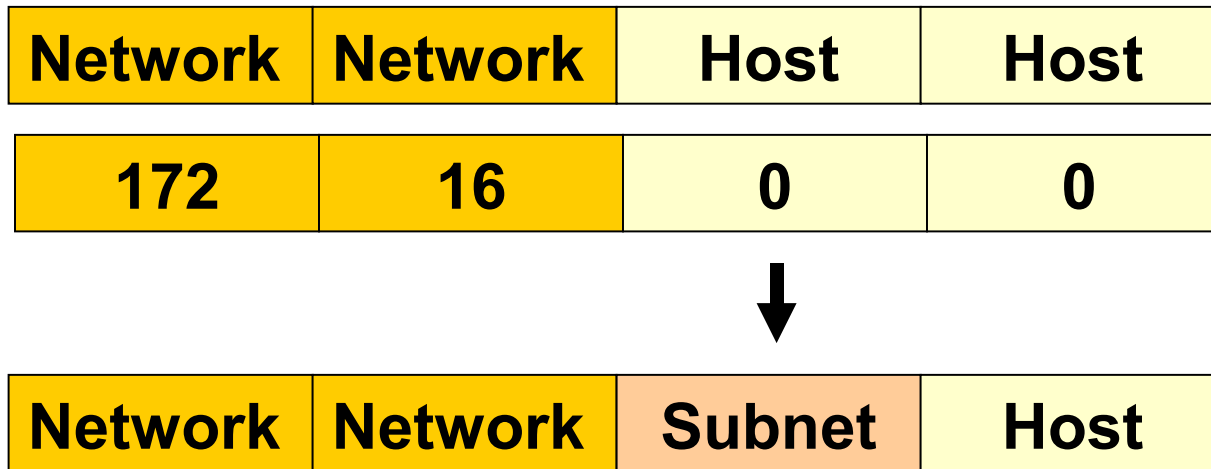
Subnets and Subnet Masks



Formalized in 1985, the subnet mask breaks a single network in to smaller pieces.

- Allows network administrators to divide their network into small networks or **subnets**.
- Advantages will be discussed later.

What is subnetting?



- Subnetting is the process of borrowing bits from the HOST bits, in order to divide the larger network into small subnets.
- Subnetting does **NOT** give you more hosts, but actually costs you hosts.
- You lose two host IP Addresses for each subnet, one for the subnet IP address and one for the subnet broadcast IP address.
- You lose the last subnet and all of it's hosts' IP addresses as the broadcast for that subnet is the same as the broadcast for the network.
- In older technology, you would have lost the first subnet, as the subnet IP address is the same as the network IP address. (This subnet can be used in most networks.)

Subnet Example

Network address 172.16.0.0 with /16 Base Network Mask

Using Subnets: **Subnet Mask 255.255.255.0 or /24**

Subnet addresses: All 0's in host portion

Network	Network	Subnet	Host
172	16	0	0
172	16	1	0
172	16	2	0
172	16	3	0
172	16	Etc.	0
172	16	254	0
172	16	255	0

**Subnets
Addresses**

**256
Subnets**

2

Subnet Example

Network address 172.16.0.0 with /16 Base Network Mask

Using Subnets: **Subnet Mask 255.255.255.0 or /24**

Network	Network	Subnet	Hosts			Broadcast
172	16	0	1	→	254	255
172	16	1	1	→	254	255
172	16	2	1	→	254	255
172	16	3	1	→	254	255
172	16	Etc.	1	→	254	255
172	16	254	1	→	254	255
172	16	255	1	→	254	255

Each subnet has 254 hosts, $2^8 - 2$

Host IP Address: 172.16.3.50

- A host of the 172.16.3.0 /24 network

With NO subnetting:

<u>Network</u>	<u>First Host</u>	<u>Last Host</u>	<u>Broadcast</u>
172.16.0.0	172.16.0.1	172.16.255.254	172.16.255.255

- 65,534 host addresses, one for network address and one for broadcast address.

Host IP Address: 172.16.3.50

- A host of the 172.16.0.0 /16 network

Host IP Address: 172.16.3.50

With subnetting:

- A host of the 172.16.3.0 /24 network

<u>Network</u>	<u>First Host</u>	<u>Last Host</u>	<u>Broadcast</u>
172.16.0.0	172.16.0.1	172.16.0.254	172.16.0.255
172.16.1.0	172.16.1.1	172.16.1.254	172.16.1.255
172.16.2.0	172.16.2.1	172.16.2.254	172.16.2.255
172.16.3.0	172.16.3.1	172.16.3.254	172.16.3.255
172.16.4.0	172.16.4.1	172.16.4.254	172.16.4.255
172.16.5.0	172.16.5.1	172.16.5.254	172.16.5.255
172.16.6.0	172.16.6.1	172.16.6.254	172.16.6.255
172.16.7.0	172.16.7.1	172.16.7.254	172.16.7.255
...			
172.16.254.0	172.16.254.1	172.16.254.254	172.16.255.255
172.16.255.0	172.16.255.1	172.16.255.254	172.16.255.255

With subnetting:

<u>Network</u>	<u>First Host</u>	<u>Last Host</u>	<u>Broadcast</u>	<u>Hosts</u>
172.16.0.0	172.16.0.1	172.16.0.254	172.16.0.255	254
172.16.1.0	172.16.1.1	172.16.1.254	172.16.1.255	254
172.16.2.0	172.16.2.1	172.16.2.254	172.16.2.255	254
172.16.3.0	172.16.3.1	172.16.3.254	172.16.3.255	254
172.16.4.0	172.16.4.1	172.16.4.254	172.16.4.255	254
172.16.5.0	172.16.5.1	172.16.5.254	172.16.5.255	254
172.16.6.0	172.16.6.1	172.16.6.254	172.16.6.255	254
172.16.7.0	172.16.7.1	172.16.7.254	172.16.7.255	254
...				
172.16.254.0	172.16.254.1	172.16.254.254	172.16.15.255	254
172.16.255.0	172.16.255.1	172.16.255.254	172.16.255.255	254

				65,024

$$\begin{aligned}
 \text{Total address} &= 256 \text{ subnets} * (256 \text{ hosts} - 2) \\
 &= 256 * 254 \\
 &= 65,024
 \end{aligned}$$

NOTE: It is common for some network administrator to not use the last subnet.

Chapter 6

IPv4 Addresses – Part 3

Topics

- Calculating the number subnets/hosts needed
- VLSM (Variable Length Subnet Masks)
- Classful Subnetting
- IPv6
- ICMP: Ping and Traceroute



Calculating the number subnets/hosts needed

Calculating the number subnets/hosts needed

2^{10}	2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1,024	512	256	128	64	32	16	8	4	2	1
Number of bits borrowed:										
10	9	8	7	6	5	4	3	2	1	
1,024	512	256	128	64	32	16	8	4	2	1
Hosts or Subnets										

172.16.1.0

255.255.255.0

Network Host

- Network 172.16.1.0/24
- Need:
 - As many subnets as possible, 60 hosts per subnet

Calculating the number subnets/hosts needed

2^{10}	2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1,024	512	256	128	64	32	16	8	4	2	1
Number of bits borrowed:					Number of hosts per subnet					
10	9	8	7	6	5	4	3	2	1	
1,024	512	256	128	64	32	16	8	4	2	1
Hosts or Subnets										

172.16.1. 0 0 0 0 0 0 0 0
 255.255.255. 0 0 0 0 0 0 0 0
 Network Host 6 host bits

- Network 172.16.1.0/24
- Need:
 - As many subnets as possible, 60 hosts per subnet

Calculating the number subnets/hosts needed

2^{10}	2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1,024	512	256	128	64	32	16	8	4	2	1
Number of bits borrowed:										
10	9	8	7	6	5	4	3	2	1	
1,024	512	256	128	64	32	16	8	4	2	1
				Hosts or Subnets				Number of subnets		

172.16.1. 0 0 0 0 0 0 0 0

255.255.255. 1 1 0 0 0 0 0 0 255.255.255.192

Network

Host

6 host bits

- Network 172.16.1.0/24
- Need:
 - As many subnets as possible, 60 hosts per subnet
- New Subnet Mask: **255.255.255.192 (/26)**
 - **Number of Hosts per subnet:** 6 bits, 64-2 hosts, **62 hosts**
 - **Number of Subnets:** 2 bits or **4 subnets**

Calculating the number subnets/hosts needed

2^{10}	2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1,024	512	256	128	64	32	16	8	4	2	1
Number of bits borrowed:										
10	9	8	7	6	5	4	3	2	1	
1,024	512	256	128	64	32	16	8	4	2	1
Hosts or Subnets										

172.16.1.0

255.255.255.0

Network Host

- Network 172.16.1.0/24
- Need:
 - As many subnets as possible, 12 hosts per subnet

Calculating the number subnets/hosts needed

2^{10}	2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1,024	512	256	128	64	32	16	8	4	2	1
Number of bits borrowed:										
10	9	8	7	6	5	4	3	2	1	
1,024	512	256	128	64	32	16	8	4	2	1
Hosts or Subnets										

Number of hosts per subnet

172.16.1. 0 0 0 0 0 0 0 0

255.255.255. 0 0 0 0 0 0 0 0

← 4 host bits

Network

Host

- Network 172.16.1.0/24
- Need:
 - As many subnets as possible, 12 hosts per subnet

Calculating the number subnets/hosts needed

2^{10}	2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1,024	512	256	128	64	32	16	8	4	2	1
Number of bits borrowed:										
10	9	8	7	6	5	4	3	2	1	
1,024	512	256	128	64	32	16	8	4	2	1
Hosts or Subnets						Number of subnets				
						Number of hosts per subnet				

172.16.1. 0 0 0 0 0 0 0 0

255.255.255. 1 1 1 1 0 0 0 0 255.255.255.240

Network

Host

4 host bits

- Network 172.16.1.0/24
- Need:
 - As many subnets as possible, 12 hosts per subnet
- New Subnet Mask: 255.255.255.240 (/28)
 - Number of Hosts per subnet: 4 bits, 16-2 hosts, 14 hosts
 - Number of Subnets: 4 bits or 16 subnets

Calculating the number subnets/hosts needed

2^{10}	2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1,024	512	256	128	64	32	16	8	4	2	1
Number of bits borrowed:										
10	9	8	7	6	5	4	3	2	1	
1,024	512	256	128	64	32	16	8	4	2	1
Hosts or Subnets										

172.16.1.0

255.255.255.0

Network Host

- Network 172.16.1.0/24
- Need:
 - Need 6 subnets, as many hosts per subnet as possible

Calculating the number subnets/hosts needed

2^{10}	2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1,024	512	256	128	64	32	16	8	4	2	1
Number of bits borrowed:										
10	9	8	7	6	5	4	3	2	1	
1,024	512	256	128	64	32	16	8	4	2	1
Hosts or Subnets							Number of subnets			

172.16.1. 0 0 0 0 0 0 0 0

255.255.255. 0 0 0 0 0 0 0 0

3 subnet bits →

Network Host

- Network 172.16.1.0/24
- Need:
 - Need 6 subnets, as many hosts per subnet as possible

Calculating the number subnets/hosts needed

2^{10}	2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1,024	512	256	128	64	32	16	8	4	2	1
Number of bits borrowed:			Number of hosts per subnet							
10	9	8	7	6	5	4	3	2	1	
1,024	512	256	128	64	32	16	8	4	2	1
Hosts or Subnets						Number of subnets				

172.16.1. 0 0 0 0 0 0 0 0

255.255.255. 1 1 1 0 0 0 0 0 255.255.255.224

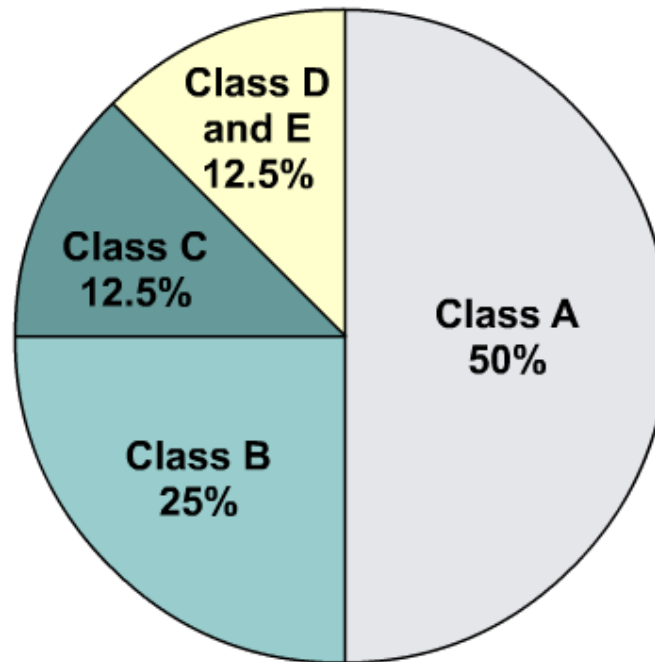
3 subnet bits →

Network

Host

- Network 172.16.1.0/24
- Need:
 - Need 6 subnets, as many hosts per subnet as possible
- New Subnet Mask: **255.255.255.224 (/27)**
 - **Number of Hosts per subnet:** 5 bits, 32-2 hosts, **30 hosts**
 - **Number of Subnets:** 3 bits or **8 subnets**

IP addressing crisis



With Class A and B addresses virtually exhausted, Class C addresses (12.5 percent of the total space) are left to assign to new networks.

- Address Depletion
- Internet Routing Table Explosion

IPv4 Addressing

2^{10}	2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1,024	512	256	128	64	32	16	8	4	2	1
Number of bits borrowed:			Number of hosts per subnet							
10	9	8	7	6	5	4	3	2	1	
1,024	512	256	128	64	32	16	8	4	2	1
					Hosts or Subnets		Number of subnets			

172.16.1. 0 0 0 0 0 0 0 0

255.255.255. 1 1 1 0 0 0 0 0 255.255.255.224

3 subnet bits

Network

Host

Subnet Mask

- One solution to the IP address shortage was thought to be the subnet mask.
- Formalized in 1985 (RFC 950), the subnet mask breaks a single class A, B or C network into smaller pieces.
- This does allow a network administrator to divide their network into subnets.
- Routers still associated an network address with the first octet of the IP address.

All Zeros and All Ones Subnets

Using the All Ones Subnet

- **There is no command to enable or disable the use of the all-ones subnet, it is enabled by default.**

```
Router(config)#ip subnet-zero
```

- **The use of the all-ones subnet has always been explicitly allowed and the use of subnet zero is explicitly allowed since Cisco IOS version 12.0.**

RFC 1878 states, "This practice (of excluding all-zeros and all-ones subnets) is obsolete! Modern software will be able to utilize all definable networks." Today, the use of subnet zero and the all-ones subnet is generally accepted and most vendors support their use, though, on certain networks, particularly the ones using legacy software, the use of subnet zero and the all-ones subnet can lead to problems.

CCO: Subnet Zero and the All-Ones Subnet

http://www.cisco.com/en/US/tech/tk648/tk361/technologies_tech_note09186a0080093f18.shtml

Long Term Solution: IPv6

- IPv6, or IPng (IP – the Next Generation) uses a 128-bit address space, yielding
340,282,366,920,938,463,463,374,607,431,768,211,456
possible addresses.
- IPv6 has been slow to arrive
- IPv6 requires new software; IT staffs must be retrained
- IPv6 will most likely coexist with IPv4 for years to come.
- Some experts believe IPv4 will remain for more than 10 years.

Short Term Solutions: IPv4 Enhancements

Class	RFC 1918 Internal Address Range	CIDR Prefix
A	10.0.0.0 to 10.255.255.255	10.0.0.0/8
B	172.16.0.0 to 172.31.255.255	172.16.0.0/12
C	192.168.0.0 to 192.168.255.255	192.168.0.0/16

Discussed in CIS 83 and CIS 185

- CIDR (Classless Inter-Domain Routing) – RFCs 1517, 1518, 1519, 1520
- VLSM (Variable Length Subnet Mask) – RFC 1009
- Private Addressing - RFC 1918
- NAT/PAT (Network Address Translation / Port Address Translation) – RFC
 - More later when we discuss TCP



VLSM (Variable Length Subnet Masks)

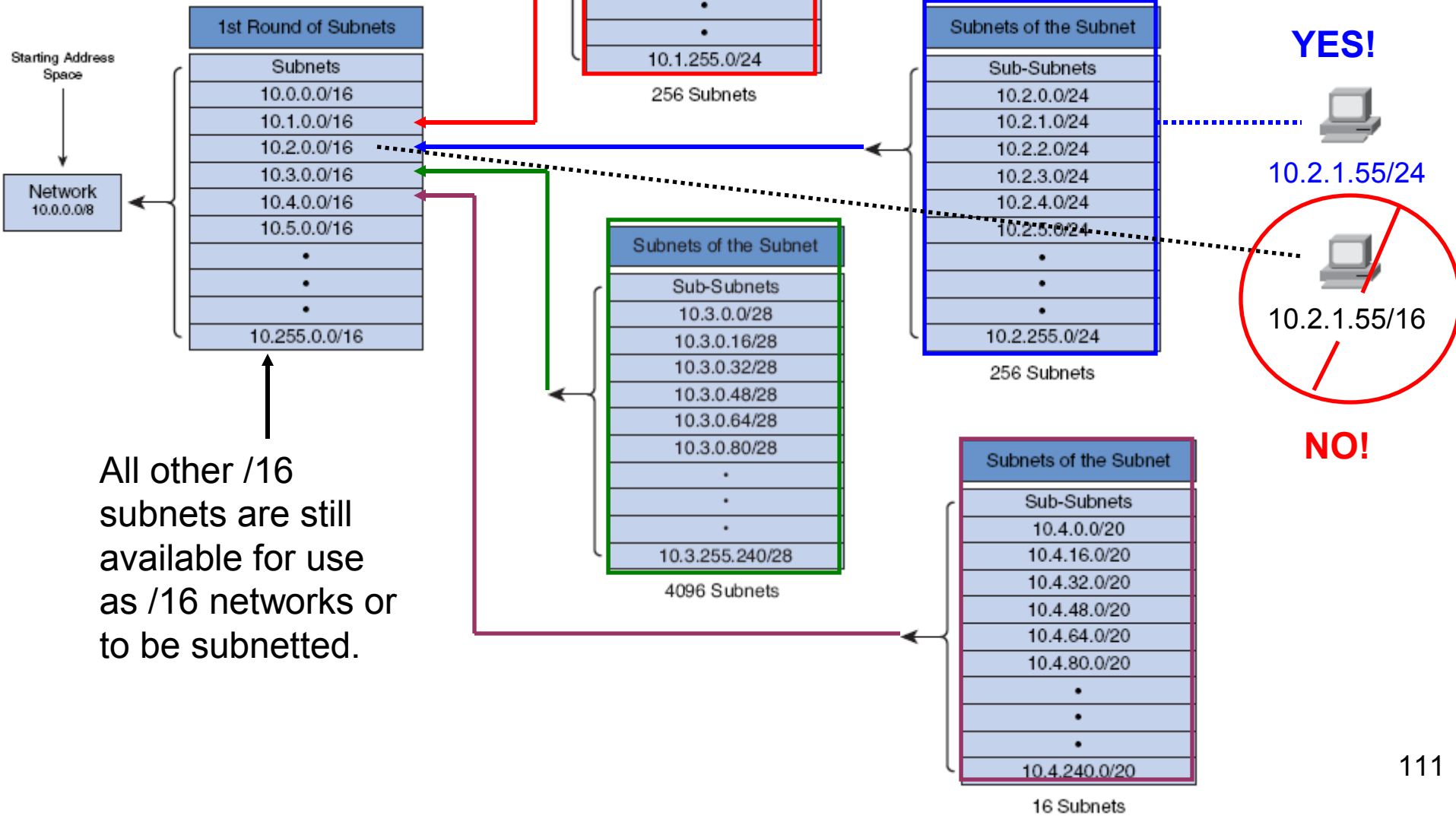
VLSM

- If you know how to subnet, you can do VLSM.
 - Example: 10.0.0.0/8
 - Subnet in /16 subnets:
 - 10.0.0.0/16
 - 10.1.0.0/16
 - 10.2.0.0/16
 - 10.3.0.0/16
 - Etc.
 - Subnet one of the subnets (10.1.0.0/16)
 - 10.1.0.0/24
 - 10.1.1.0/24
 - 10.1.2.0/24
 - 10.1.3.0/24
 - etc
- 



VLSM

Host can only be a member of the subnet. Host can NOT be a member of the network that was subnetted.



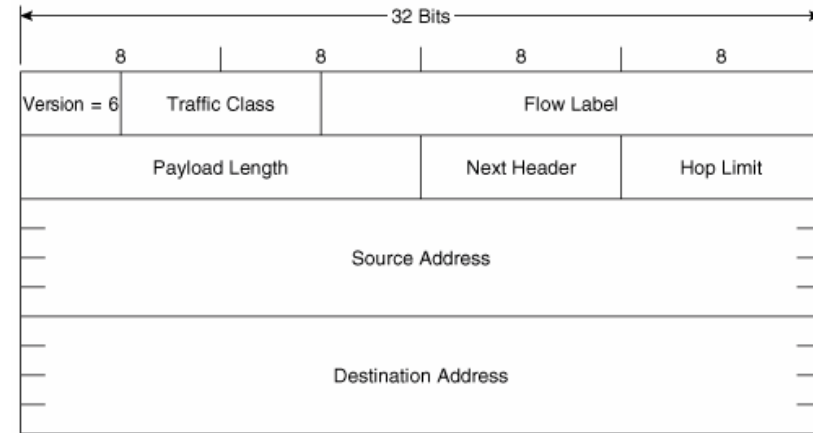
Rick Graziani
Cabrillo College

IPv6



IPv6

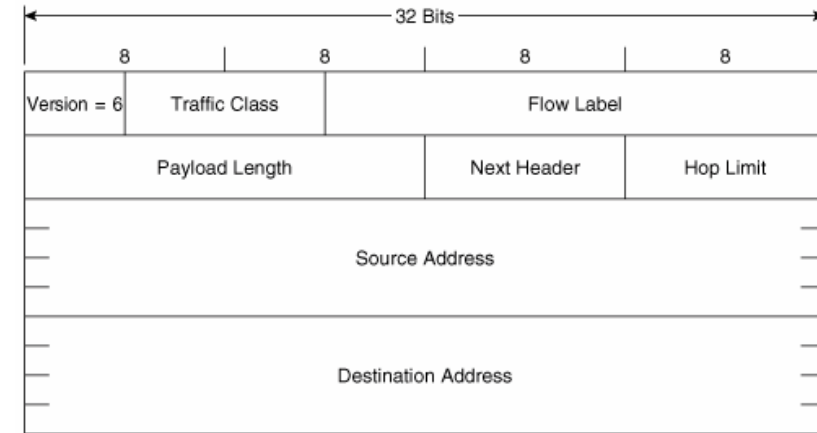
Figure 2-5. The IPv6 packet header.



- IPv6 replaces the 32-bit IPv4 address with a **128-bit address**, making **340 trillion trillion trillion IP addresses** available.
340,282,366,920,938,463,463,374,607,431,768,211,456 addresses
 - Represented by breaking them up into **eight 16-bit segments**.
 - **Each segment** is written in **hexadecimal** between 0x0000 and 0xFFFF, separated by colons.
- An example of a written IPv6 address is
3ffe:1944:0100:000a:0000:00bc:2500:0d0b

Background

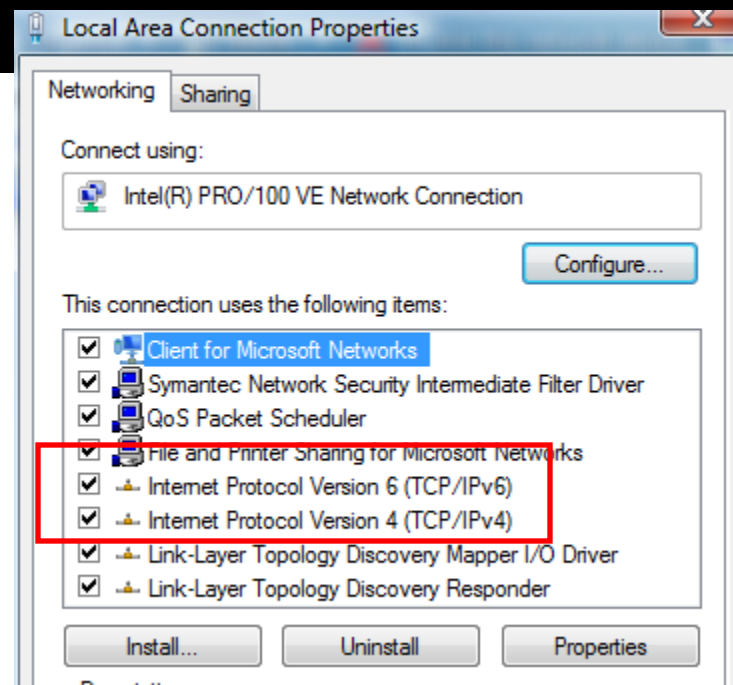
Figure 2-5. The IPv6 packet header.



- IPv4 will exist for some time, as the transition begins to IPv6.
- Other new protocols have been developed in support of IPv6:
 - Routing protocols (OSPFv3) so routers can learn about IPv6 network addresses.
 - ICMPv6

Ethernet adapter Local Area Connection:

```
Connection-specific DNS Suffix  : 
Description . . . . . : Intel(R) PRO/100 VE Network Connection
Physical Address. . . . . : 00-1B-24-04-A2-1E
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::1c00:3ea4:74ff:a8cf%8 (Preferred)
IPv4 Address. . . . . : 192.168.1.101 (Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Tuesday, April 15, 2008 7:26:06 AM
Lease Expires . . . . . : Wednesday, April 16, 2008 7:26:05 AM
Default Gateway . . . . . : 192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 201332278
DNS Servers . . . . . : 68.87.76.178
                       : 68.87.78.130
NetBIOS over Tcpip. . . . . : Enabled
```



ICMP: Ping and Trace

Ethernet Header (Layer 2)			IP Header (Layer 3)	ICMP Message (Layer 3)						Ether. Tr.
Ethernet Destination Address (MAC)	Ethernet Source Address (MAC)	Frame Type	Source IP Add. Dest. IP Add. Protocol field	Type 0 or 8	Code 0	Check- sum	ID	Seq. Num.	Data	FCS

Partial list

Type	Code	Description
0 - Echo Reply	0	Echo reply (used to ping)
1 and 2		Reserved
3 - Destination Unreachable	0	Destination network unreachable
	1	Destination host unreachable
	2	Destination protocol unreachable
	3	Destination port unreachable
	4	Fragmentation required, and DF set
	5	Source route failed
	6	Destination network unknown
	7	Destination host unknown
	8	Source host isolated
	9	Network administratively prohibited
	10	Host administratively prohibited
	11	Network unreachable for TOS
	12	Host unreachable for TOS
	13	Communication administratively prohibited
4 - Source Quench	0	Source quench (congestion control)

ICMP (Internet Control Message Protocol)

- ICMP: A Layer 3 protocol
- Used for sending messages
- Encapsulated in a Layer 3, IP packet
- Uses Type and Code fields for various messages

ICMP

Ethernet Header (Layer 2)			IP Header (Layer 3)	ICMP Message (Layer 3)						Ether. Tr.
Ethernet Destination Address (MAC)	Ethernet Source Address (MAC)	Frame Type	Source IP Add. Dest. IP Add. Protocol field	Type 0 or 8	Code 0	Check- sum	ID	Seq. Num.	Data	FCS

Unreachable Destination or Service

- Used to **notify a host that the destination or service is unreachable.**
- When a host or router receives a packet that it cannot deliver, it may send an **ICMP Destination Unreachable packet to the host** originating the packet.
- The Destination Unreachable packet will contain codes that indicate why the packet could not be delivered.

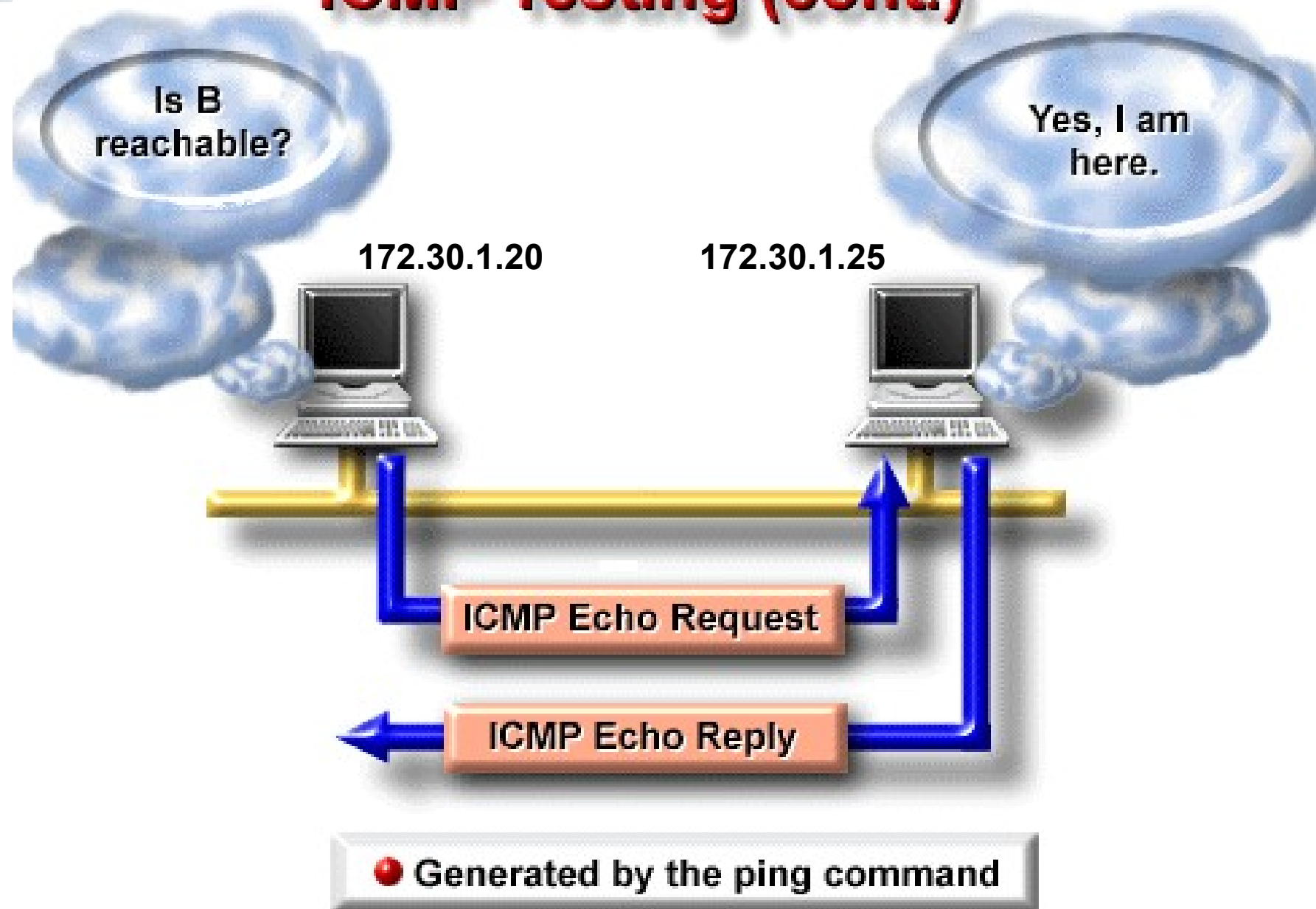
From a router:

- 0 = network unreachable – Does not have a route in the routing table
- 1 = host unreachable – Has a route but can't find host.

From a host:

- 2 = protocol unreachable
- 3 = port unreachable
 - Service is not available because no daemon is running providing the service or because security on the host is not allowing access to the service.

ICMP Testing (cont.)



Ethernet Header (Layer 2)			IP Header (Layer 3)	ICMP Message (Layer 3)						Ether. Tr.
Ethernet Destination Address (MAC)	Ethernet Source Address (MAC)	Frame Type	Source IP Add. Dest. IP Add. <u>Protocol field</u>	Type 0 or 8	Code 0	Check- sum	ID	Seq. Num.	Data	FCS

Ping

- Uses ICMP message encapsulated within an IP Packet
 - **Protocol field = 1**
- Does not use TCP or UDP

Format

- **ping** *ip address* (or **ping** <cr> for extended ping)
- **ping** *172.30.1.25*

Ethernet Header (Layer 2)			IP Header (Layer 3)	ICMP Message - Echo Request (Layer 3)						Ether. Tr.
Ethernet Destination Address (MAC)	Ethernet Source Address (MAC)	Frame Type	Source IP Add. 172.30.1.20 Dest. IP Add. 172.30.1.25 Protocol field 1	Type 8	Code 0	Check- sum	ID	Seq. Num.	Data	FCS

Echo Request

- The sender of the ping, transmits an ICMP message, “Echo Request”

Echo Request - Within ICMP Message

- Type = 8
- Code = 0

Ethernet Header (Layer 2)			IP Header (Layer 3)	ICMP Message - Echo Reply (Layer 3)						Ether. Tr.
Ethernet Destination Address (MAC)	Ethernet Source Address (MAC)	Frame Type	Source IP Add. 172.30.1.25 Dest. IP Add. 172.30.1.20 Protocol field 1	Type 0	Code 0	Check- sum	ID	Seq. Num.	Data	FCS

Echo Reply

- The IP address (destination) of the ping, receives the ICMP message, “Echo Request”
- The ip address (destination) of the ping, returns the ICMP message, “Echo Reply”

Echo Reply - Within ICMP Message

- Type = 0
- Code = 0

Ping example

```
C:\Users\rigrrazia>nslookup www.cabrillo.edu
Server:  cns.sanjose.ca.sanfran.comcast.net
Address:  68.87.76.178:53

Non-authoritative answer:
Name:     arana.cabrillo.edu
Address:  207.62.187.7
Aliases:  www.cabrillo.edu

C:\Users\rigrrazia>ping 207.62.187.7

Pinging 207.62.187.7 with 32 bytes of data:

Reply from 207.62.187.7: bytes=32 time=19ms TTL=51
Reply from 207.62.187.7: bytes=32 time=17ms TTL=51
Reply from 207.62.187.7: bytes=32 time=17ms TTL=51
Reply from 207.62.187.7: bytes=32 time=18ms TTL=51

Ping statistics for 207.62.187.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 17ms, Maximum = 19ms, Average = 17ms

C:\Users\rigrrazia>
```

Pings may fail

```
Microsoft Windows [Version 6.0.6000]
Copyright (c) 2006 Microsoft Corporation. All rights reserved.

C:\Users\rigrazia>nslookup www.cisco.com
Server: cns.sanjose.ca.sanfran.comcast.net
Address: 68.87.76.178:53

Non-authoritative answer:
Name: www.cisco.com
Address: 198.133.219.25

C:\Users\rigrazia>ping 198.133.219.25

Pinging 198.133.219.25 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 198.133.219.25:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\rigrazia>
```

Q: Are pings forwarded by routers?

A: Yes! This is why you can ping devices all over the Internet.

Q: Do all devices forward or respond to pings?

A: No, this is up to the network administrator of the device. Devices, including routers, can be configured not to reply to pings (ICMP echo requests). This is why you may not always be able to ping a device. Also, routers can be configured not to forward pings destined for other devices.

Traceroute

```
C:\WINDOWS\system32\cmd.exe

C:\>tracert www.cabrillo.edu

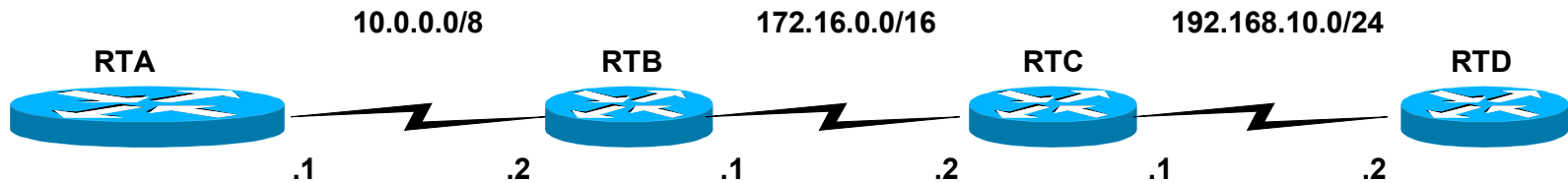
Tracing route to moose.cabrillo.edu [207.62.187.7]
over a maximum of 30 hops:

  1      1 ms      13 ms      19 ms      192.168.1.1
  2      *         *         *         Request timed out.
  3     10 ms       9 ms       9 ms       68.87.198.149
  4     14 ms      11 ms       9 ms       68.87.192.45
  5      9 ms      11 ms       9 ms       68.87.192.41
  6     11 ms      20 ms      11 ms       68.87.192.37
  7     13 ms      11 ms      11 ms       68.87.195.62
  8     12 ms      12 ms      13 ms       12.127.32.33
  9     14 ms      14 ms      13 ms       tbr1-p012301.sffca.ip.att.net [12.123.13.173]
 10     12 ms      11 ms      12 ms       12.122.81.101
 11     13 ms      14 ms      14 ms       att-gw.sea.level3.net [192.205.32.206]
 12     13 ms      14 ms      21 ms       ge-8-2.hsa1.SanJose1.Level3.net [4.68.123.136]
 13     18 ms      16 ms      17 ms       CENIC.hsa1.Level3.net [209.247.159.110]
 14     22 ms      18 ms      19 ms       dc-svl-dc1--isp-1-ge.cenic.net [137.164.22.58]
 15     21 ms      34 ms      18 ms       dc-cabrillo-ds3--svl-dc1.cenic.net [137.164.32.1
39]
 16     26 ms      22 ms      21 ms       utonium.cabrillo.edu [207.62.184.2]
 17     25 ms      31 ms      21 ms       moose.cabrillo.edu [207.62.187.7]

Trace complete.

C:\>_
```

Trace

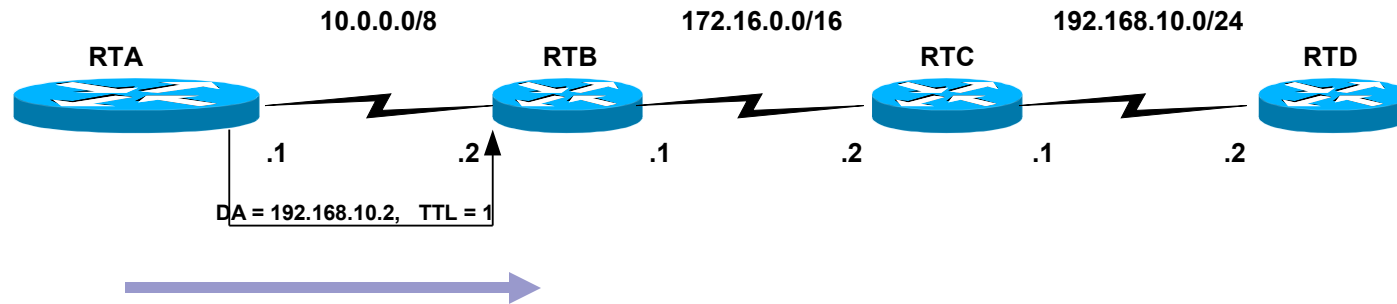


Format (trace, traceroute, tracert)

- **RTA#** *traceroute ip address*

```
RTA# traceroute 192.168.10.2
```

Trace

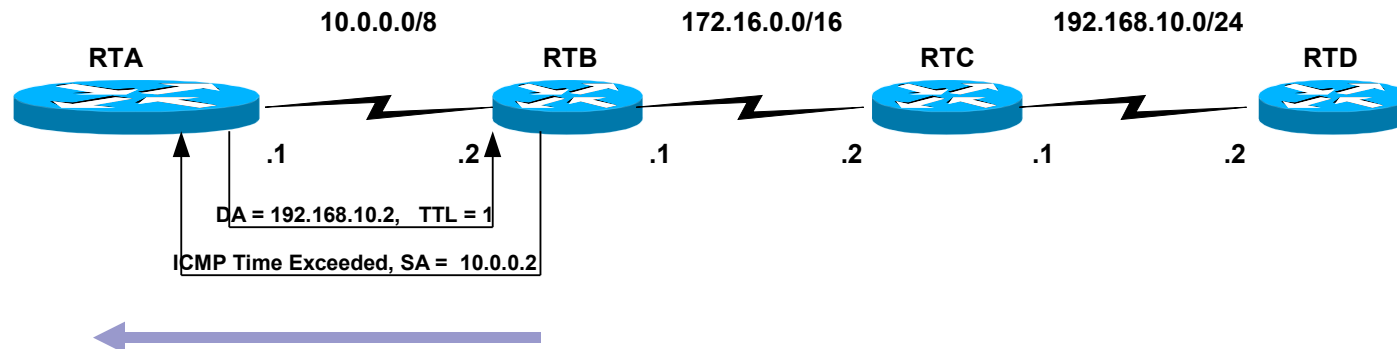


Data Link Header (Layer 2)			IP Header (Layer 3)	ICMP Message - Echo Request (trace)					UDP (Layer 4)	DataLink Tr.
Data Link Destination Address	Data Link Source Address		Source IP Add. 10.0.0.1 Dest. IP Add. 192.168.10.2 Protocol field 1 TTL 1	Type 8 Code 0	Chk sum	ID	Seq. Num	Data	DestPort 35,000	FCS

How it works (using UDP) - Fooling the routers & host!

- Traceroute uses ping (echo requests)
- Traceroute sets the TTL (Time To Live) field in the IP Header, initially to "1"

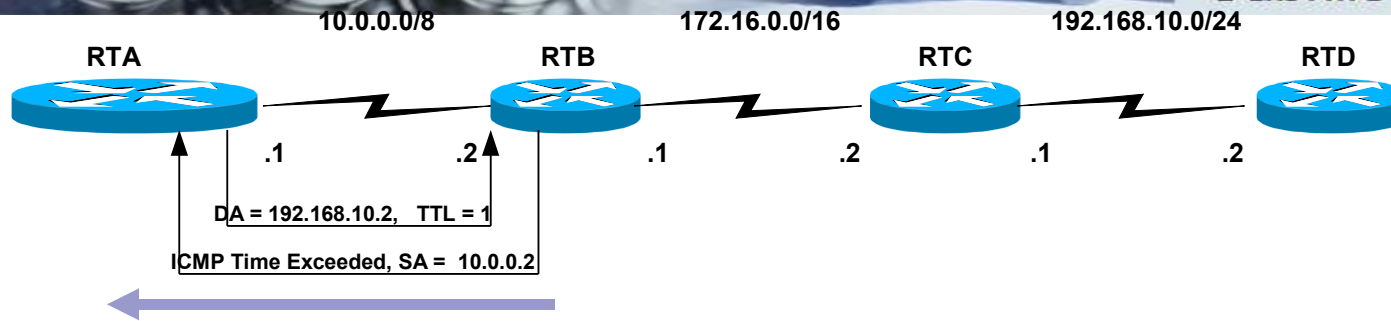
Trace



Data Link Header (Layer 2)			IP Header (Layer 3)	ICMP Message - Time Exceeded				DataLink Tr.	
Data Link Destination Address	Data Link Source Address		Source IP Add. 10.0.0.2 Dest. IP Add. 10.0.0.1 Protocol field 1	Type 11 Code 0	Chk sum	ID	Seq Nu m.	Data	FCS

RTB - TTL:

- When a router receives an IP Packet, it decrements the TTL by 1.
- If the TTL is 0, it will not forward the IP Packet, and send back to the source an ICMP "time exceeded" message.
- ICMP Message: Type = 11, Code = 0



Data Link Header (Layer 2)			IP Header (Layer 3)	ICMP Message - Time Exceeded					DataLink Tr.
Data Link Destination Address	Data Link Source Address		Source IP Add. 10.0.0.2 Dest. IP Add. 10.0.0.1 Protocol field 1	Type 11 Code 0	Chk sum	ID	Seq . Nu m.	Data	FCS

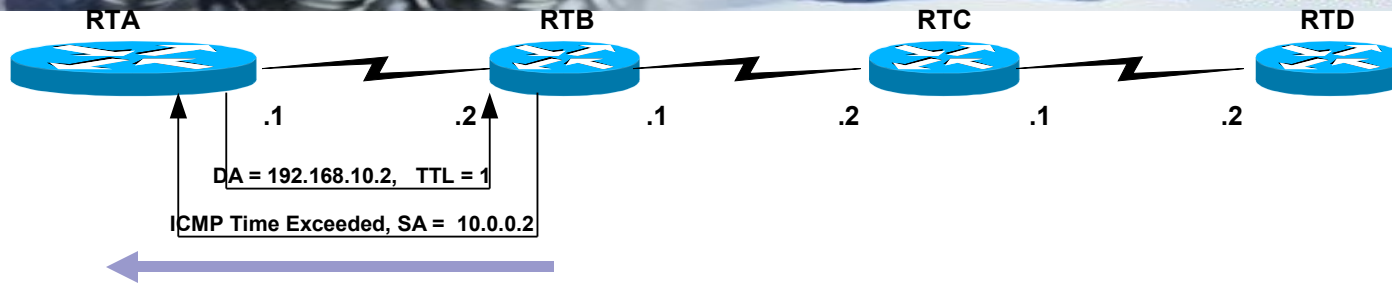
RTB

- After the traceroute is received by the first router, it decrements the TTL by 1 to 0.
- Noticing the TTL is 0, it sends back a **ICMP Time Exceeded message** back to the source, using its IP address for the source IP address.
- Router B's IP header includes its own IP address (source IP) and the sending host's IP address (dest. IP).

10.0.0.0/8

172.16.0.0/16

192.168.10.0/24



Data Link Header (Layer 2)			IP Header (Layer 3)	ICMP Message - Time Exceeded					DataLink Tr.
Data Link Destination Address	Data Link Source Address		Source IP Add. 10.0.0.2 Dest. IP Add. 10.0.0.1 Protocol field 1	Type 11 Code 0	Chk sum	ID	Seq . Nu m.	Data	FCS

RTA, Sending Host

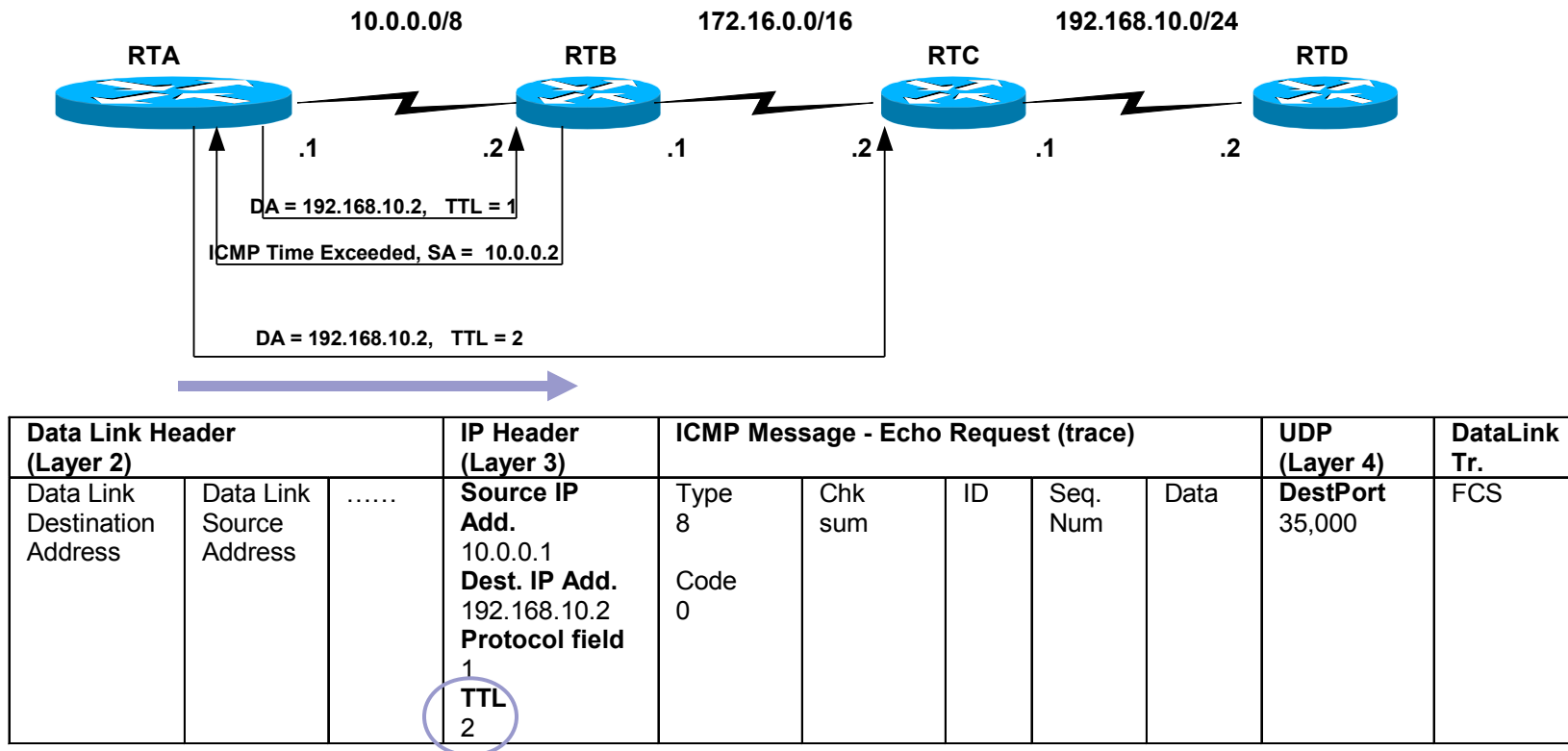
- The traceroute program of the sending host (RTA) will use the source IP address of this ICMP Time Exceeded packet to display at the first hop.

```
RTA# traceroute 192.168.10.2
```

```
Type escape sequence to abort.
```

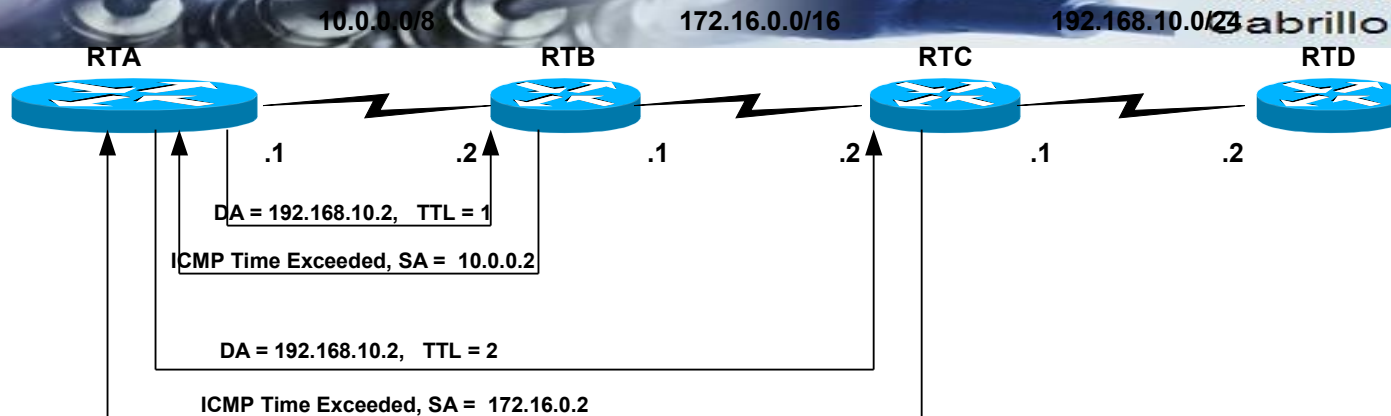
```
Tracing the route to 192.168.10.2
```

```
 1 10.0.0.2 4 msec 4 msec 4 msec
```



RTA

- The traceroute program increments the TTL by 1 (now 2) and resends the ICMP Echo Request packet.

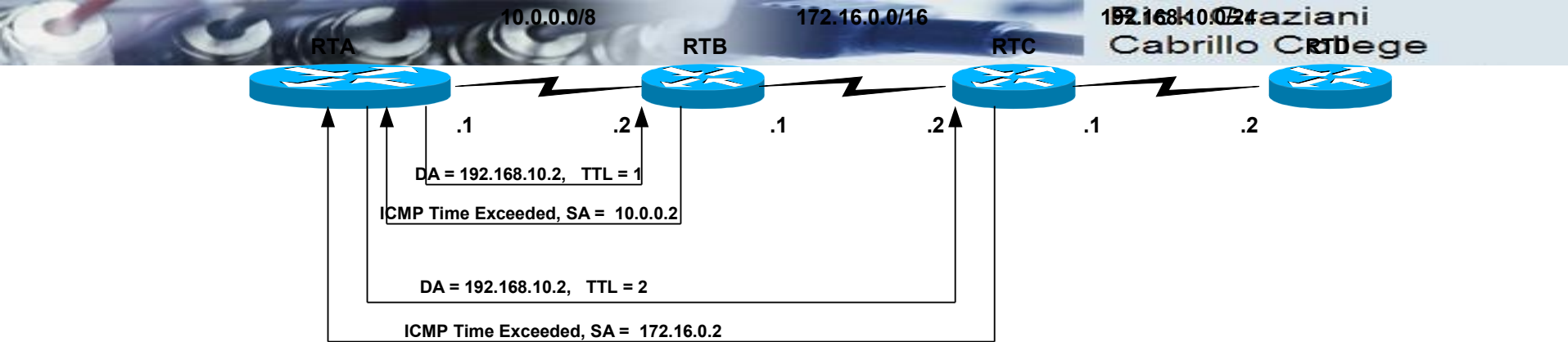


RTB

- This time RTB decrements the TTL by 1 and it is NOT 0. (It is 1.)
- So it looks up the destination ip address in its routing table and forwards it on to the next router.

RTC

- RTC however decrements the TTL by 1 and it is 0.
- RTC notices the TTL is 0 and sends back the **ICMP Time Exceeded message** back to the source.
- RTC's IP header includes its own IP address (source IP) and the sending host's IP address (destination IP address of RTA).
- The sending host, RTA, will use the source IP address of this **ICMP Time Exceeded** message to display at the second hop.



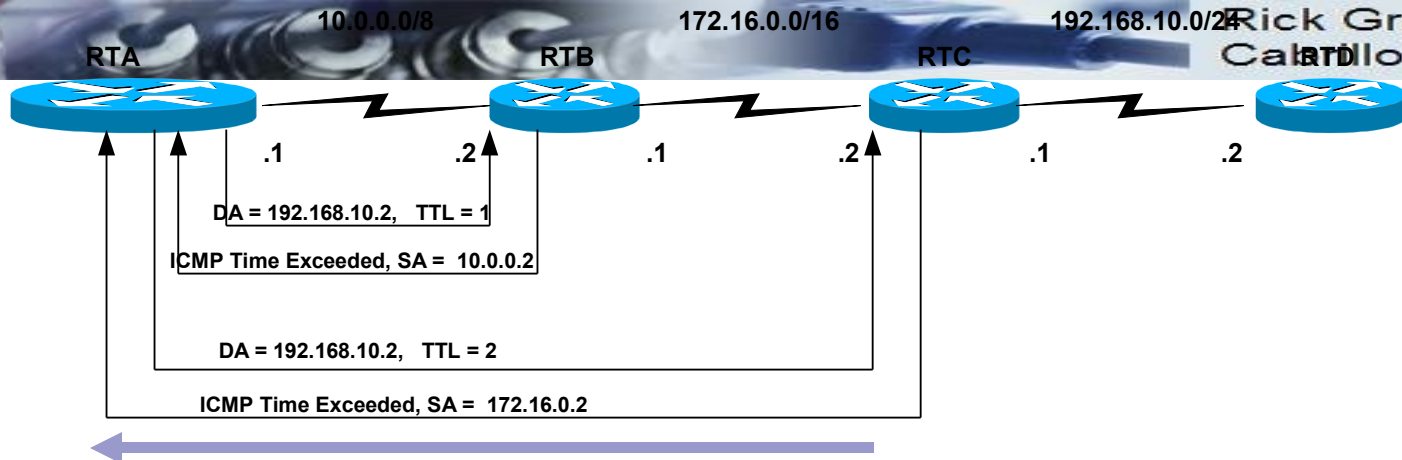
RTA to RTB

Data Link Header (Layer 2)			IP Header (Layer 3)		ICMP Message - Echo Request (trace)				UDP (Layer 4)	DataLink Tr.
Data Link Destination Address	Data Link Source Address		Source IP Add. 10.0.0.1	Dest. IP Add. 192.168.10.2	Type 8	Chk sum	ID	Seq. Num	DestPort 35,000	FCS
			Protocol field		Code 0					
			TTL 2							

RTB to RTC

Data Link Header (Layer 2)			IP Header (Layer 3)		ICMP Message - Echo Request (trace)				UDP (Layer 4)	DataLink Tr.
Data Link Destination Address	Data Link Source Address		Source IP Add. 10.0.0.1	Dest. IP Add. 192.168.10.2	Type 8	Chk sum	ID	Seq. Num	DestPort 35,000	FCS
			Protocol field		Code 0					
			TTL 1							

Data Link Header (Layer 2)			IP Header (Layer 3)		ICMP Message - Time Exceeded				DataLink Tr.
Data Link Destination Address	Data Link Source Address		Source IP Add. 172.16.0.2	Dest. IP Add. 10.0.0.1	Type 11	Chk sum	ID	Seq. Num.	FCS
			Protocol field		Code 0				
			1						



Data Link Header (Layer 2)			IP Header (Layer 3)	ICMP Message - Time Exceeded					DataLink Tr.
Data Link Destination Address	Data Link Source Address		Source IP Add. 172.16.0.2 Dest. IP Add. 10.0.0.1 Protocol field 1	Type 11 Code 0	Chk sum	ID	Seq . Nu m.	Data	FCS

The sending host, RTA:

- The traceroute program uses this information (Source IP Address) and displays the second hop.

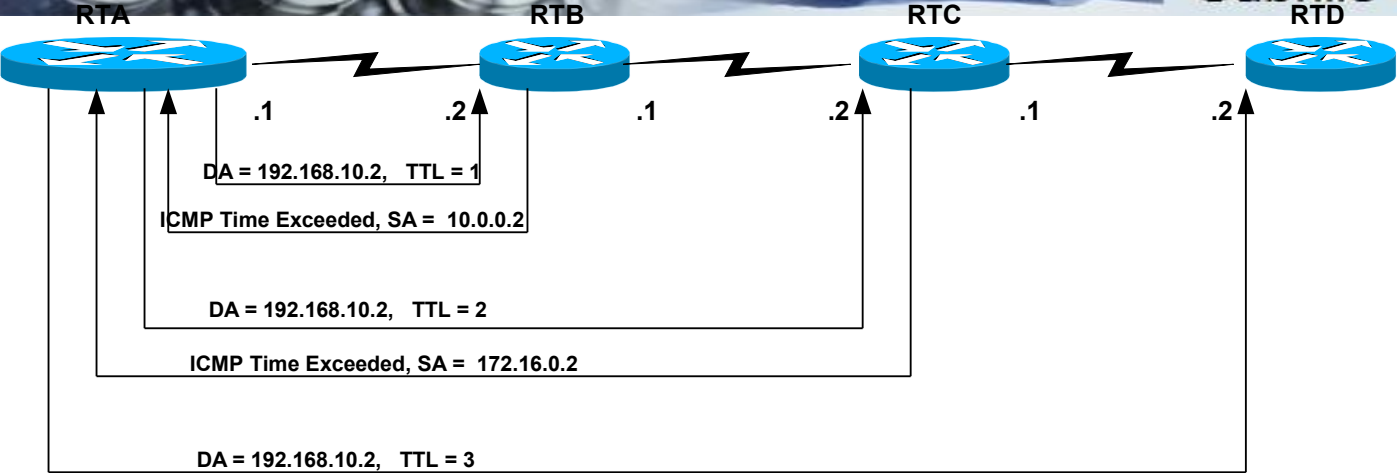
RTA# traceroute 192.168.10.2

Type escape sequence to abort.

Tracing the route to 192.168.10.2

1 10.0.0.2 4 msec 4 msec 4 msec

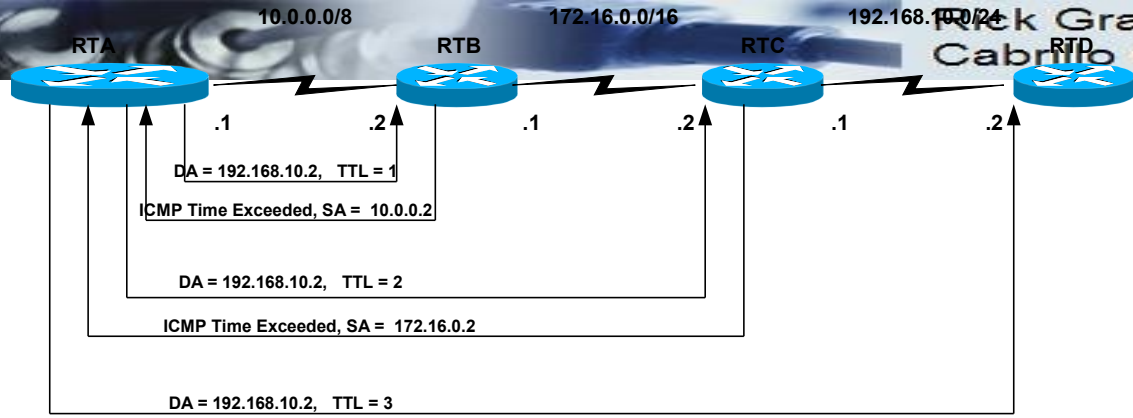
2 172.16.0.2 20 msec 16 msec 16 msec



Data Link Header (Layer 2)			IP Header (Layer 3)	ICMP Message - Echo Request (trace)					UDP (Layer 4)	DataLink Tr.
Data Link Destination Address	Data Link Source Address		Source IP Add. 10.0.0.1 Dest. IP Add. 192.168.10.2 Protocol field 1 TTL 3	Type 8 Code 0	Chk sum	ID	Seq. Num	Data	DestPort 35,000	FCS

The sending host, RTA:

- The traceroute program increments the TTL by 1 (now 3) and resends the Packet.



RTA to RTB

Data Link Header (Layer 2)			IP Header (Layer 3)	ICMP Message - Echo Request (trace)					UDP (Layer 4)	DataLink Tr.
Data Link Destination Address	Data Link Source Address		Source IP Add. 10.0.0.1 Dest. IP Add. 192.168.10.2 Protocol field 1 TTL 3	Type 8 Code 0	Chk sum	ID	Seq. Num	Data	DestPort 35,000	FCS

RTB to RTC

Data Link Header (Layer 2)			IP Header (Layer 3)	ICMP Message - Echo Request (trace)					UDP (Layer 4)	DataLink Tr.
Data Link Destination Address	Data Link Source Address		Source IP Add. 10.0.0.1 Dest. IP Add. 192.168.10.2 Protocol field 1 TTL 2	Type 8 Code 0	Chk sum	ID	Seq. Num	Data	DestPort 35,000	FCS

RTC to RTD

Data Link Header (Layer 2)			IP Header (Layer 3)	ICMP Message - Echo Request (trace)					UDP (Layer 4)	DataLink Tr.
Data Link Destination Address	Data Link Source Address		Source IP Add. 10.0.0.1 Dest. IP Add. 192.168.10.2 Protocol field 1 TTL 1	Type 8 Code 0	Chk sum	ID	Seq. Num	Data	DestPort 35,000	FCS



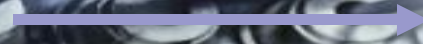
- This time RTB decrements the TTL by 1 and it is NOT 0. (It is 2.)
- So it looks up the destination ip address in its routing table and forwards it on to the next router.

RTC

- This time RTC decrements the TTL by 1 and it is NOT 0. (It is 1.)
- So it looks up the destination ip address in its routing table and forwards it on to the next router.

RTD

- RTD however decrements the TTL by 1 and it is 0.
- However, RTD notices that the Destination IP Address of 192.168.0.2 is it's own interface.
- Since it does not need to forward the packet, the TTL of 0 has no affect.



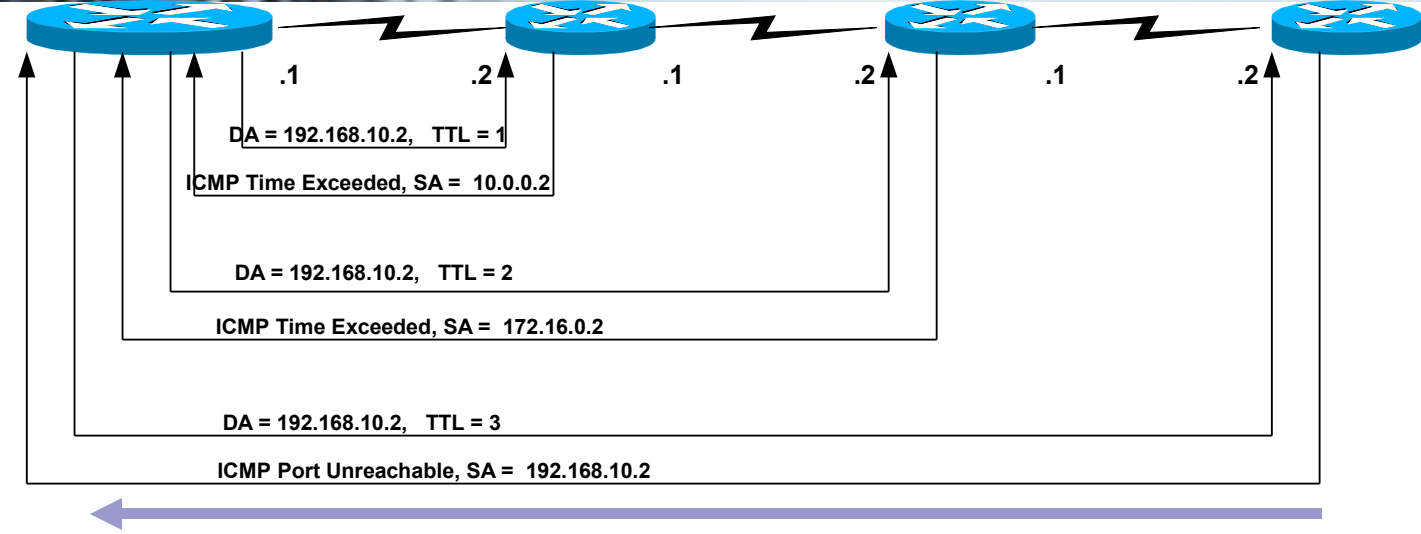
Data Link Header (Layer 2)			IP Header (Layer 3)	ICMP Message - Echo Request (trace)					UDP (Layer 4)	DataLink Tr.
Data Link Destination Address	Data Link Source Address		Source IP Add. 10.0.0.1 Dest. IP Add. 192.168.10.2 Protocol field 1 TTL 1	Type 8 Code 0	Chk sum	ID	Seq. Num	Data	DestPort 35,000	FCS



Data Link Header (Layer 2)			IP Header (Layer 3)	ICMP Message – Port Unreachable					DataLink Tr.
Data Link Destination Address	Data Link Source Address		Source IP Add. 192.168.10.2 Dest. IP Add. 10.0.0.1 Protocol field 1	Type 3 Code 3	Chk sum	ID	Seq. Num.	Data	FCS

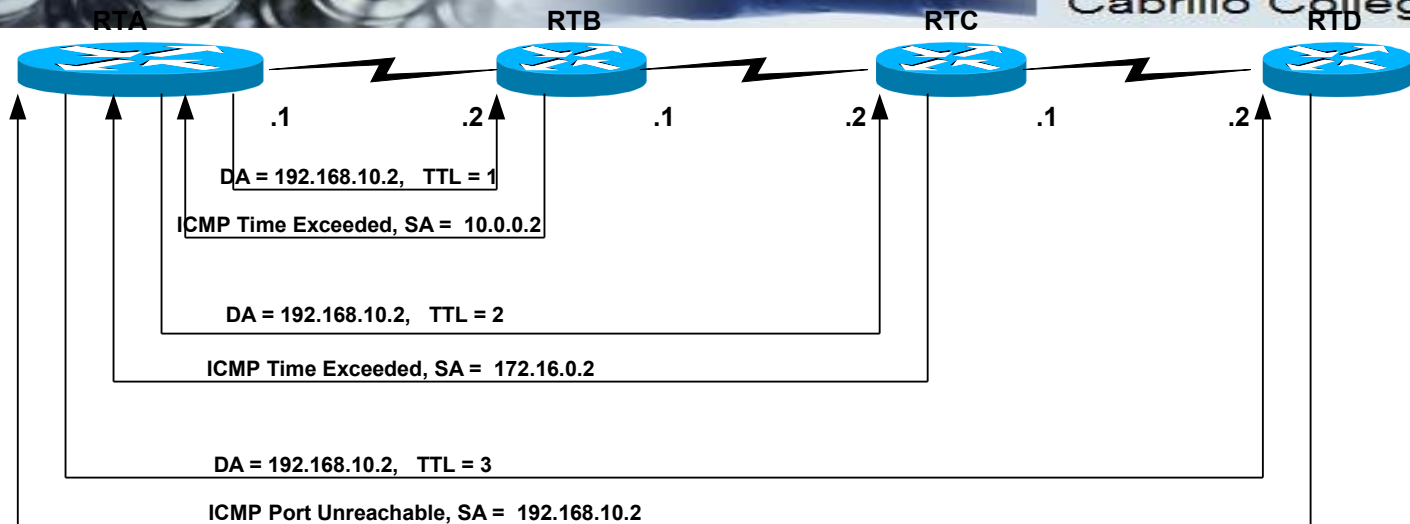
RTD

- RTD sends the packet to the UDP process.
- UDP examines the unrecognizable port number of 35,000 and sends back an **ICMP Port Unreachable message** to the sender, RTA, using Type 3 and Code 3.



Data Link Header (Layer 2)			IP Header (Layer 3)	ICMP Message – Port Unreachable					DataLink Tr.
Data Link Destination Address	Data Link Source Address		Source IP Add. 192.168.10.2 Dest. IP Add. 10.0.0.1 Protocol field 1	Type 3 Code 3	Chk sum	ID	Seq . Nu m.	Data	FCS

- ### Sending host, RTA
- RTA receives the ICMP Port Unreachable message.
 - The traceroute program uses this information (Source IP Address) and displays the third hop.
 - The traceroute program also recognizes this **Port Unreachable message** as meaning this is the destination it was tracing.



Sending host, RTA

- RTA, the sending host, now displays the third hop.
- Getting the ICMP Port Unreachable message, it knows this is the final hop and does not send any more traces (echo requests).

```
RTA# traceroute 192.168.10.2
Type escape sequence to abort.
Tracing the route to 192.168.10.2
 1 10.0.0.2 4 msec 4 msec 4 msec
 2 172.16.0.2 20 msec 16 msec 16 msec
 3 192.168.10.2 16 msec 16 msec 16 msec
```